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IDMP TEST SERIES - 2022

High Yielding PRELIMS TEST SERIES

Test – 11
Polity/Geography

Topics covered:

- Polity Revision
- Geography Revision
- Current Affairs June 2021



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DEDICATED PRELIM PROGRAM(DPP)**UPSC-CSE PRELIMINARY TEST SERIES – 2021/2022****DO NOT OPEN THIS TEST BOOKLET UNTIL YOU ARE ASKED TO DO SO****TEST BOOKLET SERIES****Test Code:****DPP****TEST BOOKLET
GENERAL STUDIES
TEST -11****E11****Time Allowed: Two Hours****Maximum Marks : 200****INSTRUCTIONS**

1. IMMEDIATELY AFTER THE COMMENCEMENT OF THE EXAMINATION, YOU SHOULD CHECK THAT THIS TEST BOOKLET DOES NOT HAVE ANY UNPRINTED OR TORN OR MISSING PAGES OR ITEMS, ETC.

2. ENCODE CLEARLY THE TEST BOOKLET SERIES CODE A, B, C OR D AS THE CASE MAY BE IN THE APPROPRIATE PLACE IN THE ANSWER SHEET.

3. You have to enter your Roll Number on the Test Booklet in the provided alongside this line. DO NOT write anything else on the Test Booklet.

4. This Test Booklet contains 100 questions. Each question is printed in English. Each item comprises four responses (answers). You will select the response which you want to mark on the Answer Sheet. In case you feel that there is more than one correct response, mark the response which you consider the best. In any case, choose ONLY ONE response for each item.

5. All items carry equal marks.

6. Penalty for wrong answers:

THERE WILL BE PENALTY FOR WRONG ANSWERS MARKED BY A CANDIDATE IN THE OBJECTIVE TYPE QUESTION PAPERS.

- I. There are four alternatives for the answer to every question. For each question for which a wrong answer has been given by the candidate, One-Third of the marks assigned to that question will be deducted as penalty.
- II. If a candidate gives more than one answer, it will be treated as a wrong answer even if one of the given answers happens to be correct and there will be same penalty as above to that question.
- III. If a question is left blank, i.e., no answer is given by the candidate, there will be no penalty for that question.

DO NOT OPEN THIS TEST BOOKLET UNTIL YOU ARE ASKED TO DO SO**TOPICS COVERED: Polity Revision, Geography Revision, Current Affairs June 2021**



Q.1) This climate is found just outside the tropics. The climate is characterized by warm moist summers and cool, dry winters. Occasionally, the penetration of cold air from the continental interiors may bring down the temperature to freezing point. Rainfall is more than moderate, between 60 cm to 150 cm. The areas are densely populated.

The above passage describes which of the following climate types?

- a) China type climate
- b) Steppe climate
- c) British type climate
- d) Siberian climate

Q.2) Consider the following statements with reference to sudden stratospheric warming:

- 1. It refers to warming in the stratosphere up to 50°C in a few days.
- 2. It might lead to an increase in ozone hole.

Which of the above statements are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.3) Arrange the following regions in the ascending order of annual rainfall received by them:

- 1. Khasi hills
- 2. Punjab
- 3. Jharkhand
- 4. Ladakh

Select the correct Answer using the code given below:

- a) 4-1-2-3
- b) 3-2-1-4
- c) 3-1-2-4
- d) 4-2-3-1

Q.4) Consider the following statements with reference to Alluvial soil:

- 1. It covers most of the Deccan Plateau including parts of Maharashtra, Madhya Pradesh, Gujarat and Tamil Nadu.
- 2. Khadar is the new alluvium and enriches soil by depositing fine silts.
- 3. Bhangar represents a system of older alluvium, composed of calcareous deposits.

Which of the above statements are correct?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1, 2 and 3

Q.5) Which of the following National capitals is closest to New Delhi in terms of latitudinal distance?

- a) Astana
- b) Bishkek
- c) Beijing
- d) Tokyo

Q.6) Consider the following pairs of local winds and the region in which they occur:

Local winds Region

- 1. Brickfielder — Southern Australia
- 2. Norte — Mexico
- 3. Buran — Mongolia
- 4. Sirocco — North Brazil

Which of the above pairs are correctly matched?

- a) 1 and 3 only
- b) 2 and 4 only
- c) 1, 2 and 3 only
- d) 1, 2 and 4 only

Q.7) It is a sedimentary rock formed out of deposits of glaciers. The Gondawana system of sediments from India is known to have its counterparts in six different landmasses (India, Falkland Island, Madagascar, Antarctica and Australia) of the Southern Hemisphere. It provides unambiguous evidence of palaeoclimates and also drifting of continents.

Which of the following rocks is best described in the above passage?

- a) Pumice
- b) Tillite
- c) Marble
- d) Quartzite

Q.8) When the saturated air mass comes across a mountain, it is forced to ascend and as it rises, it expands; the temperature falls, and the moisture is condensed in the form of rainfall. There can be large differences in the amount of rainfall on the two sides of the mountain.

Which type of rainfall is described above?

- a) Relief Rainfall
- b) Convectional Rainfall
- c) Cyclonic Rainfall
- d) None of the above

Q.9) Consider the following pairs:

Major Minerals		Application/Features
1. Pyroxene	-	Meteorites
2. Hornblende	-	Amphibole



3. Olivine - Basaltic rocks

Which of the pairs given above is/are correctly matched?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1, 2, and 3

Q.10) Consider the following statements regarding the various types of rocks:

- 1. Chalk, charcoal are examples of organically formed sedimentary rocks.
- 2. The recrystallisation of the underlying rocks is quite common in the igneous rocks.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.11) Which of the following are not the components of the sea floor spreading?

- 1. Divergent boundary interaction at the Mid-oceanic ridges.
- 2. Convergent boundary interaction at the trenches.
- 3. Increase in the age of the rocks towards the mid-oceanic ridge.
- 4. Expansion and the shrinkage of the oceans.
- 5. Occurrence of deep-seated earthquakes along the mid-oceanic ridges.

Select the correct Answer using the codes given below:

- a) 1, 3, and 5 only
- b) 3, 4, and 5 only
- c) 1, 2, and 4 only
- d) 2, 4, and 5 only

Q.12) Arrange the following tectonic plates in the South to North direction:

- 1. Cocos Plate
- 2. North American Plate
- 3. Nazca Plate

Select the correct Answer using the codes given below:

- a) 3-2-1
- b) 2-1-3
- c) 2-3-1
- d) 3-1-2

Q.13) Consider the following statements regarding cyclones:

- 1. Tropical cyclones require a pre-existing low level cyclonic circulation.
- 2. Frontal cyclones are always accompanied by air divergence above sea level.
- 3. Frontal cyclones originate on land while tropical cyclones can originate in the sea only.

Which of the statements given above is/are incorrect?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1, 2, and 3

Q.14) In relation to the soils in India, consider the following statements:

- 1. Both the alluvial and the black soil are poor in Phosphorus.
- 2. Higher temperature deprives the laterite soil of its humus content.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.15) Consider the following factors:

- 1. Decrease in the temperature of air below the dew point.
- 2. Increase in the moisture content of the air.
- 3. Increase in the temperature of the air.
- 4. Only when the value of dew point is below the freezing point.

Which of the factors given above are responsible for causing the condensation of air?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 3 and 4 only
- d) 1 and 4 only

Q.16) Consider the following pairs:

Clouds	Feature
1. Cirrus	Feathery
2. Cumulus	Cotton wool
3. Stratus	Thin and flat base
4. Nimbus	Layered

Which of the pairs given above is/are incorrectly matched?

- a) 1, 2 and 3 only
- b) 3 and 4 only
- c) 2, 3, and 4 only



d) 1, 2, and 4 only

Q.17) Which one of the following places is closest to the Strait of Hormuz?

- a) Dubai
- b) Chabahar
- c) Bandar Abbas
- d) Sharjah

Q.18) Consider the following statements regarding the tides:

- 1. Gravitational force and the Centrifugal forces are the tide-generating forces.
- 2. The tidal range increases from perihelion to aphelion.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.19) In the context of River systems of India, Consider the following statements:

- 1. Deccan Rivers fluctuate in volume as compared to the Himalayana rivers.
- 2. The Himalayan Rivers flow through deep V – shaped valleys unlike the Peninsular rivers.
- 3. Mariam La separates the source of the Brahmaputra River from the Manasarovar Lake.

Which of the statements given above is/are correct?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) All of the above

Q.20) With reference to the distribution of Volcanoes and Earthquakes, consider the following statements:

- 1. All the volcanic and the earthquake activity occur along diverging plate margins and mid-oceanic ridges.
- 2. The volcanic eruptions can cause Tsunamis to occur.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.21) Consider the following statements about Ocean Relief:

- 1. A seamount never reaches the sea surface unlike Guyots.
- 2. Ocean trenches are a result of tectonic activity, which describes the movement of the Earth's lithosphere.
- 3. The continental slope is the seaward extension of the continent from the shoreline to the continental edge.

Which of the statements given above is/are correct?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) All of the above

Q.22) Consider the following statements:

- 1. Son meta preference can lead to Gender Inequality.
- 2. Bihar has the highest total fertility rate.
- 3. Kerala was the only state that experienced an increase in TFR as per the latest National Family Health Survey.

Which of the statements given above is/are correct?

- a) 1 and 2 only
- b) 2 only
- c) 3 only
- d) All of the above.

Q.23) Consider the following statements about the Weathering and Erosion:

- 1. Both Weathering and Erosion involves displacement of eroded materials.
- 2. Rain is an agent of both Weathering and Erosion.
- 3. Temperature change is an important factor for Weathering.

Which of the statements given above is/are correct?

- a) 2 only
- b) 2 and 3 only
- c) 1 and 2 only
- d) 3 only

Q.24) Consider the following Pairs:

Reports/Indices	Meaning/Components
-----------------	--------------------

- | | |
|-----------------------------------|--|
| 1. Multidimensional Poverty Index | - Involves Education, Health and Standard of Living. |
| 2. Gender Gap Report | - Does not include Political Empowerment. |
| 3. Gender Inequality Index | - Lower values in the GII indicate worse achievements. |

Select the correct Answer using the code below.

- a) 1 and 2 only
- b) 2 and 3 only



- c) 3 only
- d) 1 only

Q.25) Consider the following statements:

1. Primary activities include animal husbandry, forestry, fishing, manufacturing, mining and quarrying.
2. In the northern hemisphere, Frontogenesis happens in an anti-clockwise direction.
3. The shape of bays along a coastline can magnify the intensity of tides.

Which of the statements given above is/are incorrect?

- a) 1 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) All of the above

Q.26) With reference to the Salinity of the Oceans, consider the following statements:

1. Evaporation and Ocean Currents are the determinants of Ocean Salinity.
2. The Salinity of the Red Sea is greater than that of the Indian Ocean.
3. Ocean Salinity can provide ideas about the Precipitation Patterns.

Which of the statements given above is/are correct?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) All of the above

Q.27) With reference to Soils, consider the following statements:

1. Khadar is more porous than the Bhangar soils.
2. The Black soils are the result of intense leaching due to rains.
3. Excessive irrigation in a dry climatic condition results in the deposition of salt on the top layer of the soil.

Which of the statements given above is/are correct?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) All of the above

Q.28) Consider the following statements with regards to frontogenesis:

1. If a warm air mass moves towards the cold air mass, the contact zone turns into a cold front.

2. Occluded front is formed by air mass sinking to the land surface.
3. The air fronts bring abrupt changes in temperature and cause the air to rise to form clouds and cause precipitation.

Select the correct Answer using the codes below.

- a) 1 and 2 only
- b) 2 only
- c) 1 and 3 only
- d) 3 only

Q.29) Which of the following forms the major components of Volcanic Smog (Vog)?

1. Ozone and Nitrogen dioxide (NO₂)
2. Sulfur dioxide (SO₂)
3. Aerosols
4. Gaseous Hydrocarbons

Select the correct Answer using the code given below.

- a) 1, 2 and 3 only
- b) 1 and 4 only
- c) 2 and 3 only
- d) 1, 2, 3 and 4

Q.30) Consider the following statements, with regards to the "Exosphere layer":

1. In the exosphere layer, atoms and molecules of atmospheric gases constantly collide with each other.
2. Exosphere layer displays a dizzy glow of ultraviolet radiation scattered by hydrogen atoms.
3. The International Space Station (ISS) orbits within the exosphere or below it.

Select the correct Answer using the codes below.

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1, 2, and 3

Q.31) Which of the following factors are responsible for the variations in insolation at the surface of the Earth?

1. Revolution of the earth around the sun
2. Rotation of earth on its axis
3. The angle of inclination of the sun's rays
4. Length of the day
5. Transparency of the atmosphere

Select the correct Answer using the code given below:

- a) 1, 2, 3, and 4 only
- b) 2, 3, 4, and 5 only



- c) 1, 2, and 3 only
- d) 1, 2, 3, 4 and 5

Q.32) Consider the following statements with regards to Jet streams:

1. Jet streams travel in the tropopause which is the boundary between the turbulent troposphere and the calm, cold stratosphere.
2. Jet Streams are faster in winter when the temperature differences between tropical, temperate, and polar air currents are greater.

Which of the above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) None

Q.33) Oceans distant from deserts or with limited accessibility to dust-carrying winds from deserts often have limited primary productivity. This is due to

- a) Lack of iron nutrient supplies
- b) Presence of kelp forests (macroalgae) in such areas
- c) Absence of a Photic Zone
- d) Warm water temperature

Q.34) With regards to Temperature Inversion, Consider the following statements:

1. Limited diffusion of dust and smoke are the likely effects of the inversion of temperature.
2. Inversion of temperature can increase the concentration of air pollutants in low-lying valleys.
3. Inversion of temperature causes atmospheric instability.

Which of the given statements is/are correct?

- a) 1 only
- b) 1 and 2 only
- c) 2 and 3 only
- d) All of the above

Q.35) Match the following:

List I

- A. Clouds having a feathery appearance, formed at high altitudes
- B. Clouds look like cotton wool, formed at mid altitudes
- C. Black or dark gray clouds formed very

List II

- 1. Cumulus Cloud
- 2. Cirrus Cloud
- 3. Nimbus Cloud

near to the surface

D. Layered clouds

covering large

portions of the sky

Select the correct Answer using the code given below:

A B C D

- (a) 1 2 3 4
- (b) 2 3 1 4
- (c) 2 1 3 4
- (d) 4 1 3 2

Q.36) Which of the following statements is/are correct about the different Oceanic zones?

1. The mesopelagic zone is referred to as the Twilight zone because of the scarce amount of sunlight.
2. Benthic zone is the area where the maximum amount of sun's radiation falls.
3. Bathypelagic zone is referred to as the midnight zone in which no light penetrates.

Select the correct Answer using the code given below:

- a) 2 only
- b) 3 only
- c) 1 and 3 only
- d) 1 and 2 only

Q.37) Consider the following pairs with reference to names of rocks and their types:

Name of rocks Type of rocks

- 1. Gabbro — Plutonic rocks
- 2. Sandstone — Sedimentary rocks
- 3. Limestone — Metamorphic rocks

Which of the above pairs are correctly matched?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1, 2 and 3

Q.38) These volcanoes are characterised by eruptions of cooler and more viscous lavas than basalt. These volcanoes often result in explosive eruptions. The material coming out of the volcano accumulates in the vicinity of the vent opening leading to formation of layers.

Which of the following types of the volcanoes is being referred to in the above passage?

- a) Shield volcanoes
- b) Composite volcanoes
- c) Caldera



d) Mid-Ocean ridge volcanoes

Q.39) Consider the following statements with reference to Mountains:

1. Himalayas and Western Ghats are examples of fold mountains.
2. Mount Merapi is an example of a volcanic mountain.
3. The Deccan plateau is an example of residual mountains.

Which of the above statements are correct?

- a) 1 and 2 only
- b) 1 and 3 only
- c) 2 and 3 only
- d) 1, 2 and 3

Q.40) Consider the following statements with reference to Mass movements:

1. Earthflow is the movement of water-saturated clayey or silty earth materials down low-angle terraces.
2. Debris slide is the slipping of one or several units of rock debris with a backward rotation.
3. Mudflow is the rapid movement of water-saturated weathered materials without vegetation cover.

Which of the above statements are correct?

- a) 1 and 2 only
- b) 2 only
- c) 2 and 3 only
- d) 1 and 3 only

Q.41) Which of the following river erosions can be seen in the mountain course of a river?

1. River capture
2. Meanders
3. Cataracts
4. Ox-bow lakes

Select the correct Answer using the code given below:

- a) 1 and 2 only
- b) 1 and 3 only
- c) 2 and 4 only
- d) 2 and 3 only

Q.42) Consider the following statements:

1. The Masai tribes are mainly found in the African countries bordering Mediterranean Sea.
2. Lalang are the prominent features of the forests found in the tropical areas.

Which of the statements given above is/are incorrect?

- a) 1 only
- b) 2 only

c) Both 1 and 2

d) Neither 1 nor 2

Q.43) Consider the following pairs:

Strait	Associated country
--------	--------------------

- | | |
|------------------|-----------|
| 1. Bab al Mandab | Djibouti |
| 2. Gibraltar | Portugal |
| 3. Malacca | Singapore |
| 4. Bosphorus | Turkey |

Which of the pairs given above is/are correctly matched?

- a) 1, 2, and 3 only
- b) 2, 3, and 4 only
- c) 1, 3, and 4 only
- d) 1, 2, and 4 only

Q.44) Consider the following statements:

1. The process of frontogenesis is usually associated with that of the air mass over homogeneous surfaces.
2. The point of convergence of easterlies around the equator is referred to as trade winds.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.45) Consider the following statements:

1. While ozone prevents harmful UV radiation from reaching the surface of the earth, carbon dioxide prevents terrestrial radiation.
2. While oxygen is found only upto a certain height in the atmosphere, water vapour and carbon dioxide are found in large quantities till the exosphere.
3. The concentration of water vapor in the atmosphere is decreased polewards and also with the increase in the altitude.

Which of the statements given above is/are correct?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1, 2, and 3

Q.46) Which one of the statements given below is least appropriate regarding the geomorphological processes?

- a) Epeirogenic processes are a part of Diastrophism, which is an endogenic process.



b) Ground water can also act as an agent for the occurrence of erosion.

c) Alternate dry and wet conditions promote the process of salt crystallization.

d) Process of Exfoliation is independent of the temperature changes.

Q.47) With reference to Urbanisation, consider the following statements:

1. Bandicoot is an automated robot which would help end the practice of manual scavenging.
2. Under the Saansad Adarsh Gram Yojana, the MPs can use the MPLAD funds for developing Adarsh Gram.
3. TULIP is a platform to connect graduates to Urban Local Bodies to create solutions for our cities.

Which of the statements given above is/are correct?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) All of the above

Q.48) Consider the following statements about atolls:

1. Atoll is a roughly circular oceanic reef system surrounding a large central lagoon.
2. Addu atoll is the southernmost atoll of Maldives
3. Addu atoll lies north to the equator.

Which of the statements given above is/are correct?

- a) 1 and 2 only
- b) 1 and 3 only
- c) 2 and 3 only
- d) 1, 2 and 3

Q.49) Horn of Africa comprises which of the following countries?

- a) Djibouti, Ethiopia, Sudan, Eritrea
- b) Ethiopia, Somalia, Djibouti, Eritrea
- c) Djibouti, Somalia, Eritrea, Egypt
- d) Somalia, Tanzania, Kenya, Uganda

Q.50) Consider the following statements with reference to Indian Monsoon:

1. Indian Ocean Dipole is the difference between temperature of eastern and western Indian Ocean.
2. Early onset of monsoon has a direct impact on the quantum of rainfall received during the season.
3. A negative Indian Ocean Dipole is associated with warmer sea surface temperatures in eastern Indian

Ocean and cooler temperatures in western Indian Ocean.

Which of the statements given above is/are correct?

- a) 1 only
- b) 1 and 2 only
- c) 1 and 3 only
- d) 1, 2 and 3

Q.51) Consider the following statements:

1. The position of this atmospheric pressure belt varies with the apparent movement of the Sun.
2. It is a zone of convergence because the winds flowing from sub-tropical high-pressure belts converge here.
3. This belt extends from the equator to 10°N and 10°S latitudes.

Which of the following atmospheric belts is described in the above statements?

- a) Subtropical high-pressure belt
- b) Sub polar low-pressure belt
- c) Equatorial low-pressure belt
- d) High pressure belt

Q.52) Consider the following statements:

1. The Gumti River connects Meghalaya and Bangladesh.
2. The Bharathapuzha river is a tributary of the Cauvery River.
3. The Dzukou valley is located in the state of Sikkim.

Which of the statements given above is/are correct?

- a) 3 only
- b) 1 and 2 only
- c) 2 and 3 only
- d) None of the above.

Q.53) Consider the following statements:

1. Aksai Chin connects Tibet with Xinjiang.
2. The territories of India, Bhutan and China intersect in the Doklam Plateau, which is adjacent to the Chumbi Valley.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.54) Consider the following factors:

1. The Matching of Continents
2. Placer deposits
3. Distribution of fossils
4. Tillites



Which of the above is/are considered as evidence in support of the continental drift theory?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1,2 and 4 only
- d) 1,2,3 and 4

Q.55) Consider the following pairs:

Straits	Connects
1. Formosa	Sea of okhotsk and Sea of Japan
2. Tartary	South China sea and East China sea
3. Otranto	Adriatic and Ionian Sea
4. Ten-degree channel	Andaman Sea and the Bay of Bengal

Which of the pairs given above is/are correctly matched ?

- a) 1 and 2 only
- b) 1 and 4 only
- c) 2 and 3 only
- d) 3 and 4 only

Q.56) Which of the following are associated with the Indian ocean?

- 1. Chagos Archipelago
- 2. Christmas Island
- 3. Gulf of Tonkin
- 4. Grand Banks

Select the correct Ans: using the code given below:

- a) 1 and 2 only
- b) 2 and 3 only
- c) 3 and 4 only
- d) 1,2,3 and 4

Q.57) Consider the following pairs:

Volcanoes	Country
1. Popocatépetl	México
2. Anak Krakatau	Hawaii
3. Mauna Loa	Indonesia
4. Ojos del salado	Chile

Which of the pairs given above is/are correct ?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 3 and 4 only
- d) 1 and 4 only

Q.58) Among the following group of countries, which one is a part of the Arctic circle?

- a) USA, Canada, Estonia, Finland
- b) Finland, Denmark, Norway, Sweden

- c) USA, Russia, Belarus, Estonia
- d) Russia, Canada, Estonia, Denmark

Q.59) consider the following pairs:

Plate boundaries features	Associated
1. Divergent boundary	Folding, faulting, trenching
2. Convergent boundary	Rift valley, sea floor spreading
3. Transform boundary	Faulting, Plates slide past each other

Which of the pairs given above is/are incorrectly matched?

- a) 1 and 2 only
- b) 1 and 3 only
- c) 2 and 3 only
- d) None of the above

Q.60) Consider the following pairs:

Hydro power projects	River
1. The Somasila Hydro Power Project	Godavari River
2. The Chutak Hydro Power Project	Chenab River
3. The Nimoo Bazgo Power Project	Indus River

Which of the pairs given above is/are correctly matched?

- a) 1 and 2 only
- b) 3 only
- c) 1 and 3 only
- d) 1, 2 and 3 only

Q.61) Which of the following seas share its maritime boundary with Taiwan?

- 1. The East China Sea
- 2. The South China Sea
- 3. The Philippine sea

Select the correct Answer using the code given below:

- a) 1 and 3 only
- b) 2 only
- c) 2 and 3 only
- d) 1, 2 and 3

Q.62) Which of the following statements with reference to 'Coral Triangle' is/are not correct?

- 1. It is a marine area located in the western Pacific Ocean.
- 2. Lakshadweep group of islands is a part of this coral triangle.

3. Coral Triangle Initiative is launched by UNEP to address the growing threats to the Coral Triangle

Select the correct Answer using the code given below.

- (a) 1 and 2 only
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

Q.63) Consider the following statements:

- 1. Carbon dioxide and water vapour are found only up to 90 km from the surface of the earth.
- 2. Inert gases constitute about 75% of the atmosphere.

Which of the above statements is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.64) Seabed 2030 Project, sometimes seen in the news is related to -

- a) Removing plastics from the seabed.
- b) Coral Reefs conservation.
- c) Ocean Floor Mapping.
- d) Project to improve the life of aquatic species.

Q.65) Which of the following industries are classified as "Red Category" industries:

- 1. Nuclear power plants
- 2. Sugar industries
- 3. Firecrackers industries
- 4. Tea processing industries
- 5. Glass manufacturing industries

Select the correct Ans: using the code given below:

- a) 1 and 2 only
- b) 1, 2 and 3 only
- c) 1, 2, 4 and 5 only
- d) 1,2,3,4 and 5

Q.66) Consider the following Pairs:

Lakes	States
1. Ropar	- Punjab
2. Vembanad	- Kerala
3. Renuka	- Manipur
4. Loktak	- Himachal Pradesh

Which of the pairs given above is/are correctly matched?

- a) 1 and 2 only
- b) 1, 2 and 3 only
- c) 1, 2 and 4 only

- d) 2,3 and 4 only

Q.67) Which of the given geographic bodies are endorheic?

- 1. Pangong Tso
- 2. Aral Sea
- 3. Caspian Sea
- 4. Sambhar lake

Select the correct Answer using the code given below:

- a) 1 and 2 only
- b) 1 and 3 only
- c) 1, 3 and 4 only
- d) 1, 2, 3 and 4

Q.68) Consider the following statements with reference to Super cyclones:

- 1. Cyclones that have a maximum wind speed of 120 knots are classified as Super cyclones.
- 2. The recent Cyclone Tauktae was recognised as a super cyclone.
- 3. Cyclone Tauktae made its landfall between Sagar islands and Hatiya islands.
- 4. Cyclone Tauktae was named by Myanmar.

Which of the statements given above is/are correct?

- a) 1, 2 and 3 only
- b) 2 and 3 only
- c) 1 and 4 only
- d) 1, 2, 3 and 4

Q.69) Consider the following pairs of protected areas and the state in which they are located:

Protected area	State
1. Barnadi wildlife sanctuary	— Odisha
2. Dehing-Patkai elephant reserve	— Assam
3. Pakhui tiger reserve	— West Bengal

Which of the above pairs are correctly matched?

- a) 1 and 2 only
- b) 2 only
- c) 3 only
- d) 1 and 3 only

Q.70) Consider the following statements with reference to Gharial:

- 1. Currently their major population is only found in India and Nepal.
- 2. They are categorized as endangered in IUCN red list but scheduled in Schedule I of Wildlife Protection Act, 1972.



3. Satkosia gorge of Mahanadi is the southernmost limit of gharials' home range in India.

Which of the above statements are correct?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1, 2 and 3

Q.88) Consider the following pairs with reference to species and their protection status:

Species	Protection Status
1. Saltwater crocodile	— Critically endangered
2. Olive Ridley turtle	— Vulnerable
3. Pygmy Hog	— Endangered

Which of the above pairs is/are correctly matched?

- a) 1 only
- b) 1 and 2 only
- c) 2 and 3 only
- d) 3 only

Q.89) Which of the following statements with reference to Shanghai Cooperation Organisation is correct?

- a) It was established immediately after the end of the Cold war to maintain peace in the region.
- b) Out of the 8 member countries, 5 are central Asian countries.
- c) Both India and Pakistan became members of the SCO in the same year.
- d) None of the above

Q.90) Which of the following organizations/institutions has recently released the Global Peace Index, 2021?

- a) Reporters without borders
- b) UNICEF
- c) World Economic Forum
- d) None of the above

Q.91) The term 'Bharitalasuchus tapani', recently seen in news, is related to:

- a) An early human species
- b) A geological period in the history of Indian subcontinent
- c) A carnivorous reptile species
- d) Pleistocene-era rock paintings in Indonesia

Q.92) Consider the following statements regarding the glacial lakes of Ganga basin:

- 1. Uttarakhand accounts for around 93% of the glacial lakes.
- 2. Yamuna sub-basin contains the highest count of the glacial lakes.
- 3. Lakes with more than 5 hectares of the area account for around 75% of the total number of lakes.

Which of the statements given above is/are incorrect?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1, 2, and 3

Q.93) Consider the following pairs:

1. Raimona National Park	:	Assam
2. Vishdhari Wildlife Sanctuary	:	Gujarat
3. Netravali Wildlife Sanctuary	:	Goa
4. Papikonda National Park	:	Andhra Pradesh

Which of the pairs given above is/are correctly matched?

- a) 1 and 3 only
- b) 2 and 3 only
- c) 2 and 4 only
- d) 1 and 4 only

Q.94) Consider the following statements:

- 1. SAGE portal is aligned with the objectives mentioned under Article 41 of the Constitution.
- 2. Surakshit Hum Surakshit Tum Abhiyan was launched for the aspirational districts only.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.95) In the context of GST, which of the statements given below is/are incorrect?

- 1. The GST on gold and unworked semi-precious stones is subject to the slab of 5%.
- 2. Value Added Tax in the liquor levied by the States have been subsumed under the GST.

Select the correct answer using the codes given below:

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2



Q.96) Consider the following statements regarding the Non-Banking Financial Companies- Micro-finance Institutions in India:

1. The current definition of Microfinance borrowers classifies them into rural, urban, and semi-urban categories.
2. Presently, the RBI regulates all the entities registered under the microfinance sector under a single set of norms.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.97) Which one of the statements given below is most appropriate regarding the LEAF Coalition?

- a) It is an initiative of the Least Developed and Island Countries that aims to arrest the global warming induced sea level rise.
- b) It is a public-private partnership that aims to mobilize finance for the protection of tropical forests.
- c) It is a United Nations led initiative that encourages afforestation in the Least Developed Countries.
- d) It is an initiative of Libya, Egypt, Algeria, and France to prevent the pollution in the Mediterranean Sea.

Q.98) Which of the following statements is/are correct about the Pradhan Mantri Awas Yojana (URBAN)?

1. Economically weaker sections and low-income groups are the only beneficiaries under the Pradhan Mantri Awas Yojana (URBAN).
2. It is being implemented as a Centrally Sponsored Scheme except for the component of credit linked subsidy which is implemented as a Central Sector Scheme.

Select the correct answer using the codes given below.

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.99) Lakshadweep comes under which of the following jurisdiction of the High Court ?

- a) Bombay High Court
- b) Kerala High Court
- c) Karnataka High Court
- d) None of the Above

Q.100) Which of the following provisions are provided under the article 324 of the constitution of India regarding the composition of the election commission?

1. The Election Commission shall be a three-member body, consisting of the chief election commissioner and two election commissioners.
2. The president may appoint regional commissioners after consultation with the chief election commissioner of India.
3. The conditions of service and tenure of office of the election commissioners and the regional commissioners shall be determined by the Parliament.

Select the correct answer using the codes given below.

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) None of the Above



Q.1) Ans: A

Exp:

About Warm Temperate Eastern Margin (China type) Climate

- This type of climate is found on the **eastern margins of continents** in warm temperate latitudes, just **outside the tropics**. It is, in fact, the climate of most parts of **China** —a **modified form of monsoonal climate**.
- It has comparatively **more rainfall** than **Mediterranean** climate.
- The climate is typified by a **warm moist summer and a cool, dry winter**. The mean monthly temperature varies between 40°F to 78°F and is strongly modified by **maritime influence**.
- Occasionally, the penetration of cold air from the continental interiors may bring down the temperature to freezing point. Though **frosts are rare** they occasionally occur in the colder interiors.
- **Rainfall** is more than moderate, anything from **60 cm to 150 cm**. The areas which experience this climate are very **densely populated**.

Reference:

G C Leong

Q.2) Ans: A

Exp:

- **Statement 1 is correct:** The term **sudden stratospheric warming** refers to what is observed in the **stratosphere**, a **rapid warming** (up to about **50°C** in just a couple of days), between 10 km and 50 km above the earth's surface. This is because in some years, the winds in the **polar vortex temporarily weaken**, or even **reverse to flow from east to west**. The **cold air** then **descends very rapidly** in the polar vortex and this causes the **temperature** in the stratosphere to **rise very rapidly**.
- **Statement 2 is incorrect:** The **Antarctic ozone hole** forms during the **Southern Hemisphere's late winter** as the returning Sun's rays **start ozone-depleting reactions**. These reactions involve **chemically active forms of chlorine and bromine** derived from man-made compounds and occur on the surfaces of cloud particles that form in **cold stratospheric layers**. In **warmer temperatures fewer polar stratospheric clouds** form and they don't persist as long, thus **limiting the ozone-depletion process**. During the sudden

stratospheric warming in 2019, as per **NASA and US National Oceanic and Atmospheric Administration (NOAA)** reports, the **ozone hole** over the Antarctic was the **smallest** observed since 1982.

Reference:

<https://indianexpress.com/article/explained/explained-how-the-ozone-layer-hole-over-arctic-closed-6381835/>

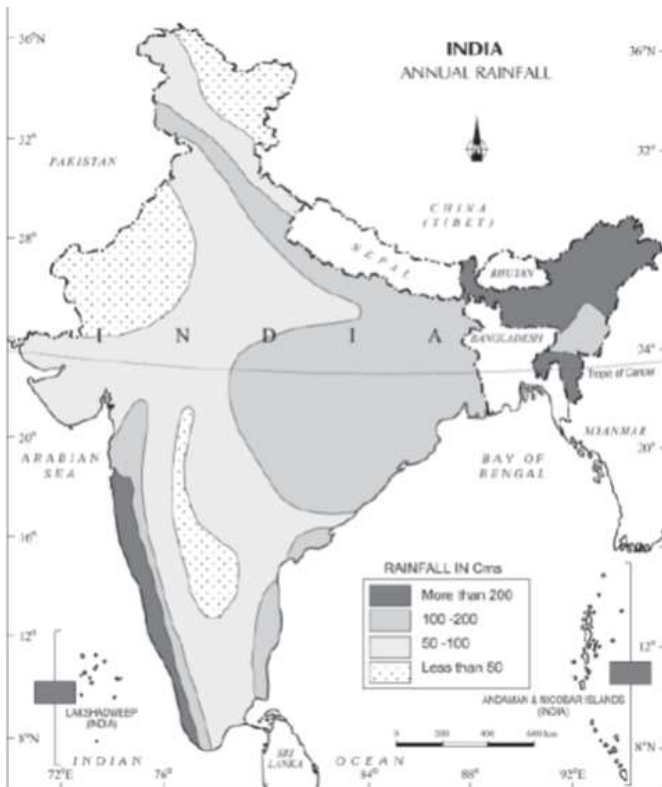
<https://www.metoffice.gov.uk/weather/learn-about/weather/types-of-weather/wind/sudden-stratospheric-warming>

Q.3) Ans: D

Exp:

Option D is correct

- The **highest rainfall** occurs along the west coast, on the **Western Ghats**, as well as in the **sub-Himalayan areas** in the **northeast** and the **hills of Meghalaya**. Here the rainfall **exceeds 200 cm**. In some parts of **Khasi and Jaintia hills**, the **rainfall exceeds 1,000 cm**.
- Rainfall between **100-200 cm** is received in the **northeastern Peninsula** covering **Odisha, Jharkhand, Bihar, eastern Madhya Pradesh, northern Ganga plain** along the **sub-Himalayas** and the **Cachar Valley** and **Manipur**.
- **Western Uttar Pradesh, Delhi, Haryana, Punjab, Jammu and Kashmir, eastern Rajasthan, Gujarat** and the **Deccan Plateau** receive **rainfall between 50-100 cm**.
- Parts of the **Peninsula**, especially in **Andhra Pradesh, Karnataka and Maharashtra, Ladakh** and most of **western Rajasthan** receive rainfall **below 50 cm**.



Reference: NCERT

Q.4) Ans: B

Exp:

- **Statement 1 is incorrect:** Black soil covers most of the Deccan Plateau which includes parts of Maharashtra, Madhya Pradesh, Gujarat, Andhra Pradesh and some parts of Tamil Nadu. Alluvial soils are widespread in the northern plains and the river valleys. These soils cover about 40 per cent of the total area of the country. In the Peninsular region, they are found in deltas of the east coast and in the river valleys.
- **Statement 2 and 3 are correct:** In the Upper and Middle Ganga plain, two different types of alluvial soils have developed, viz. Khadar and Bhangar. Khadar is the new alluvium and is deposited by floods annually, which enriches the soil by depositing fine silts. Bhangar represents a system of older alluvium deposited away from the flood plains. Both the Khadar and Bhangar soils contain calcareous concretions (Kankars). These soils are more loamy and clayey in the lower and middle Ganga plain and the Brahmaputra valley. The sand content decreases from the west to east.

Reference: NCERT Page no. 69

Q.5) Ans: D

Exp:

Astana: Astana is the capital of Kazakhstan and located at 51.16°N latitude.

Bishkek: Bishkek is the capital of Kyrgyzstan and located at 42.87°N latitude.

Beijing: Beijing is the capital of China and located at 39.90°N latitude.

Tokyo: Tokyo is the capital of Japan and located at 35.67°N latitude.



Q.6) Ans: C

Exp:

- **Pair 1 is correct:** The Brickfielder is a strong, hot, dry and dusty wind in southern Australia. The Brickfielder usually occurs during summer and is mainly affecting southeast Australia's states of Victoria and New South Wales.
- **Pair 2 is correct:** Norte is a strong, cold wind that blows from the northeast in Mexico and on the shores of the Gulf of Mexico. It results from a wintertime outbreak of cold air from the north.
- **Pair 3 is correct:** Buran A strong cold northeasterly wind creating extreme blizzard conditions during winter. The Buran occurs most frequently in Mongolia and Siberia and Buran winds are strong and full of ice and snow. This Buran is much feared in the region and can be persistent and closes passes for weeks as it blows. Over the tundra it is also known as Purga. In



Alaska this severe **north-easterly wind** is known as **Burga** bringing snow and ice pellets.

- **Pair 4 is incorrect:** The term **Sirocco** is an all-inclusive name given to **hot** and subsequently **humid southeast to southwest winds** originating as hot, dry desert-air over **Northern Africa**, flowing northward into the southern Mediterranean basin.

Reference:

<https://www.weatheronline.co.uk/reports/wind/The-Buran-or-Burga.htm>

<https://www.chicagotribune.com/news/ct-xpm-2001-02-07-0102070063-story.html>

<https://www.weatheronline.co.uk/reports/wind/Brickfielder.htm>

<https://www.weatheronline.co.uk/reports/wind/The-Sirocco.htm>

Q.7) Ans: B

Exp:

- **Tillite** is the **sedimentary rock** formed out of **deposits of glaciers**. The **Gondwana system** of sediments from India is known to have its **counterparts in six different landmasses** of the Southern Hemisphere. At the base the system has thick tillite indicating **extensive and prolonged glaciation**. Counter parts of this succession are found in **Africa, Falkland Island, Madagascar, Antarctica and Australia** besides **India**. The glacial tillite provides **unambiguous evidence of palaeoclimates** and also **drifting of continents**.
- **Pumice** is a type of **extrusive volcanic rock**, produced when **lava** with a **very high content of water** and **gases** is discharged from a volcano. When this lava **cools and hardens**, the result is a very light rock material filled with tiny bubbles of gas. It is an **igneous rock**.
- **Marble** is a **metamorphic rock** composed of **recrystallized carbonate minerals**, most commonly **calcite or dolomite**.
- **Quartzite** is a **metamorphic rock** formed when **quartz-rich sandstone or chert** has been exposed to high temperatures and pressures.

Reference:

NCERT

<https://mineralseducationcoalition.org/minerals-database/pumice/>

Q.8) Ans: A

Exp:

- **Relief or Orographic rainfall:** When the **saturated air mass** comes across a **mountain**, it is forced to **ascend** and as it **rises**, it **expands**; the **temperature falls**, and the **moisture is condensed**. The chief characteristic of this sort of rain is that the **windward slopes** receive **greater rainfall**. After giving rain on the windward side, when these winds reach the **other slope**, they **descend**, and their **temperature rises**. Then their **capacity** to take in **moisture increases** and hence, these **leeward slopes** remain **rainless and dry**. The area situated on the leeward side, which gets less rainfall, is known as the **rain-shadow area**.
- **Convictional rainfall:** When the **air is heated** it becomes **light and rises** up in **convection currents**. As it **rises**, it **expands** and **loses heat** and consequently, **condensation** takes place and **cumulus clouds** are formed, leading to **rainfall**.
- **Cyclonic rainfall:** Cyclonic Rainfall is **convictional rainfall** on a **large scale**. The precipitation in a **tropical cyclone** is of **conventional** type while that in a **temperate** cyclone is because of **frontal activity**.

Reference: NCERT

Q.9) Ans: D

Exp:

- **Minerals** are the naturally occurring organic and inorganic substances having the definite composition and the physical properties. Oxygen is the most abundant element present in the Earth's crust. The source of minerals is the magma in the interior of the Earth. Metallic minerals are the ones that contain metals in them, e.g., the gold, silver, iron, aluminum, zinc, etc. while the non-metallic minerals lack the mineral content, for example, Sulphur, phosphate, nitrate, etc. Feldspar, Quartz, Pyroxene, Amphibole, Mica, Olivine are some of the examples of the major minerals.
- **Pyroxene** is a green or black color mineral forming around 10 percent of the earth's crust. It is mainly found in meteorites and is composed of calcium, aluminum, magnesium, iron and silica.



- **Amphiboles** comprise 7 percent of the Earth's crust and are mainly used in the asbestos industry. It is found in green or black color and is mainly composed of Aluminum, Silica, calcium, iron, and Magnesium. Hornblende is also a form of Amphibole.
- **Olivine** is a greenish mineral found in the basaltic rocks. It is used in jewelry and made of Magnesium, iron, and silica.

Source: ncert 12th Chapter 5

Q.10) Ans: A

Exp:

- **Statement 1 is correct:** **Rocks** are the combination of one or two minerals and are found in various varieties. There are different kinds of rocks found in the earth's crust namely **igneous rocks, metamorphic rocks** and the **sedimentary rocks**. **Igneous rocks** are also called the primary rocks and are formed with the solidification of the magma and the lava that comes out from the surface of the earth. **Sedimentary rocks** are formed due to the action of several exogenous forces. They are usually deposited in layers and classified into three major groups: Mechanically formed like limestone, shale, sandstone, etc. Organically formed like chalk, limestone, coal, etc. Chemically formed like limestone, chert, potash, etc.
- **Statement 2 is incorrect:** **Metamorphic rocks** are formed with the change in the internal structure of the rocks on account of the forces of pressure, temperature and volume that a rock is subjected to. In the process of formation of the Metamorphic rocks, rocks undergo recrystallisation and reorganization of materials within the original rocks. Granite, slate, quartzite, marble, etc. are the examples of the metamorphic rocks.

Source: NCERT physical 12th page 42

Q.11) Ans: B

Exp:

- The concept of **sea floor spreading** was proposed by **Hess** and seeks to provide an explanation to **Alfred Wegner's theory of Continental Drift**.
- As per Hess' concept of the sea floor spreading, the continuous eruptions occurring at the crest of the oceanic ridges are responsible for the pushing of the oceanic crust on either side leading to the spreading of the oceanic crust. However, since the expansion of one does not accompany the shrinkage of another ocean, therefore he also explained the consumption of the oceanic crust along the trenches. Hence, as per the concept of the sea floor, there occurs the divergent boundary interactions occur at the mid-oceanic ridges.
- The volcanic eruptions at the mid-oceanic ridges are responsible for the formation of the younger crust near the oceanic ridge. The convergent boundary interaction occurs at the boundary of the oceanic and the continental crust. It leads to the subduction of the ocean crust at the trenches.
- The rocks are younger near the mid-oceanic ridges and their age increases as one moves away from the ridge towards the trench. There does not occur the shrinkage of the oceanic crust as explained by Hess. The newly formed oceanic crust at the crest is consumed in the trenches. Earthquakes are shallower along the mid-oceanic ridge and deep-seated earthquakes are usually witnessed along the continental margins surrounding the Pacific ocean forming the Pacific Ring of Fire.

Source: Ncert physical 12th pg 33

Q.12) Ans: D

Exp:

- As per the Plate Tectonic theory, the earth is composed of the number of Major and minor plates which move horizontally over the asthenosphere of the earth. The Major Plates are the: **Antarctica Plate, North American Plate, South American Plate, Pacific Plate, India-Australia Plate, Africa Plate, Eurasian Plate.**
- **Cocos plate, Nazca Plate, Caribbean Plate, Philippine Plate, Caroline Plate, Arabian Plate**

are the examples of the minor plates. The Nazca Plate is found between the South American and the Pacific Plates The Cocos Plate is found between the Central American and the Pacific Plates. **he North American Plate is located** To the northeast of Cocos plate.

- The correct order of the given plates in the South to North direction is the Nazca Plate, Cocos Plate, and the North American Plate. Hence, 3-1-2 is the correct order.

Source: NCERT physical 12th page 35

Q.13) Ans: B

Exp:

- **Statement 1 is correct:** The **tropical cyclones** are the violent storms that originate along the oceans in the tropical areas. They are accompanied by heavy rainfall and huge destruction due to the movement of the fast blowing winds. They are called by different names in different regions, namely Willy-Willy in the Western Australia, Hurricanes in the Atlantic, Typhoons in the South China Sea, and Cyclones in the Indian Ocean. They require a certain pre condition for their formation which includes:
 - Larger sea with temperature around 27 degree Celsius
 - Presence of Coriolis force
 - A pre-existing weak low pressure condition
 - Upper divergence of air above the sea surface.
- **Statement 2 is incorrect:** **Front cyclones** or the **middle latitude** or **extra tropical cyclones** are formed due to the movement of the air mass or the frontal system. These cyclones can originate either on the land or the sea and require a clear front for their formation. These cyclones occur with the interaction of the hot and the cold air fronts. The presence of upper divergence of the air over the sea system is not a necessary precondition of the frontal cyclones. It is a tropical cyclone that requires such a condition.
- **Statement 3 is incorrect:** **Frontal cyclones** can originate both on the land and the sea and

move from the west to east direction while the **tropical cyclones** can originate in the sea only and move in the east to west direction.

Source: NCERT 12th physical pg 95

Q.14) Ans: C

Exp:

- **Statement 1 is correct:** **Soil** is a mixture of rock debris and organic matter and is an essential component for the growth of plants. A number of factors affect the formation of soil namely the relief, parent material, climate, and vegetation, other life-forms and time. Soils are classified as *Urvara* and *Usara* which are the fertile and the sterile soils respectively. On the basis of their genesis, color, composition, and location, soils can be classified into Alluvial soil, Black soil, Red and Yellow soil, Laterite soil, Arid soil, Saline soil, Peaty soils, and Forests soils.
 - **Alluvial soils** are found in the northern plains and the river valleys. They are depositional soils and cover around 40 percent of the total area of the country. They are rich in potash and poor in phosphorus.
 - **Black soil** is mainly found in the Deccan Plateau and is called the **Regur** or the **Black Cotton soil**. They swell and become sticky when wet and shrink when dry. In the dry season they develop cracks and are self ploughing. They lack phosphorus, nitrogen, and organic matter. Hence, both the alluvial and the black soil are poor in phosphorus.
- **Statement 2 is correct:** **Laterite soils** are found in the regions of high temperature and heavy rainfall. As a result of intense heating, the silica in the soil is leached away and soil rich in iron oxide and aluminum is left. Owing to higher temperature, the huge bacterial activity of the soil removes the soil of its humus content. They are poor in organic matter, potash, nitrogen, and calcium and are not suitable for cultivation. They are found in the areas of Tamil Nadu, Andhra Pradesh, and Kerala and



are suitable for the production of tree crops like Cashew.

Source: ncert 12th Indian geo page 69-70

Q.15) Ans: A

Exp:

- **Humidity** refers to the water vapor present in the air. The air containing moisture to its full capacity is the saturated air and the temperature at which the saturation of the air occurs is called the **dew point**.
- The increase in temperature of the air increases its water holding capacity and with greater movement of the air, greater is the rate of evaporation. **The conversion of the water vapor present in the air into water is called Condensation** and it occurs with the loss of heat.
- Condensation is favourable when the temperature of the air is reduced to a value below the dew point at a constant volume. Increase in the moisture content of the air through evaporation also increases the chances for condensation. The most important condition for the condensation to occur is the decrease in air temperature.
- Hence, increase in temperature causes evaporation. Condensation can occur when the dew point is either above or below the freezing point. When the dew point is above the freezing point, condensation can occur in the form of **dew**. When the dew point is below the freezing point, the condensation occurs in the form of **frost**.

Source Ncert 12th physical pg 98

Q.16) Ans: B

Exp:

- **Clouds** are formed at a certain distance above the surface of the earth and are composed of the mass of minute water droplets or crystals of ice that are formed by the condensation of water vapor in air. Depending upon their height, density, expanse, transparency, clouds are classified into different types: **cirrus clouds, cumulus clouds, Stratus clouds, Nimbus clouds.**

- **Cirrus Clouds** are thin and white in color. They have a feathery appearance and are formed at higher altitudes.
- **Cumulus clouds** have a cotton wool appearance and occur at a height of 4000 to 7000 m. They are scattered and have a flat base. These clouds occur at elevations less than the cirrus clouds.
- **Stratus clouds** are layered clouds covering large portions of the sky. These clouds are generally formed either due to loss of heat or the mixing of air masses with different temperatures.
- **Nimbus clouds** are black or grey in appearance and are opaque to the rays of the sun owing to higher density. They generally occur very near to the surface of the earth.

Source: Ncert 12th physical page 99

Q.17) Ans: C

Exp:

- **The Strait of Hormuz** is a narrow strip of water connecting the Persian Gulf and the Gulf of Oman. It is present between Oman and Iran. **Bandar Abbas** in Iran is closest to the strait of Hormuz. It is present in the narrow region of Persian Gulf in the southern part of Iran. **Dubai** is situated in the Persian Gulf and is an important city of the UAE. **Sharjah** is a city in the UAE situated in the Persian Gulf and to the northeast of Abu Dhabi, its capital. **Chabahar** is a city situated in the Gulf of Oman. It is a strategic port city located on the Makran coast of Iran. Chabahar opens to the Indian ocean.

Source: Atlas

Q.18) Ans: A

Exp:

- **Statement 1 is correct:** **Tides** refer to the periodic rise and fall of the sea level either once or twice in a day due to the attraction by the sun and the moon. The gravitational force of the moon is primarily responsible for the occurrence of tides. Besides, the gravitational force of the sun and the centrifugal forces are also responsible for the formation of the tides.



Hence, the gravitational attraction of the moon and the centrifugal forces are together responsible for the generation of tides and are referred to as the tide generating forces.

- **Statement 2 is incorrect:** Perihelion is the time when the earth is closest to the sun on 3rd January. During this time, the tidal ranges are unusually high and unusually low. Aphelion is when the earth is farthest from the sun on 4th July. It exhibits less than average tidal ranges. Hence, while the tidal ranges are high during Perihelion, they are low during Aphelion. Therefore, the tidal ranges decrease from the Perihelion to Aphelion.

Source: NCERT 12th physical pg 121, 122

Q.19) Ans: D

Exp:

- **Statement 1 is correct:** The Himalayan Rivers are formed by melting snow and glaciers and therefore, continuously flow throughout the year. During the monsoon months, Himalayas receive very heavy rainfall and rivers swell, causing frequent floods. The Deccan Rivers on the other hand are rain fed and therefore fluctuate in volume. Many of these are non-perennial.
- **Statement 2 is correct:** The Himalayan rivers flow through deep V – shaped valleys called gorges. Gorges are formed by the interplay of several geological processes, including erosion, tectonic processes such as vertical uplift and cavern collapse. Erosion by the resident body of water is usually the primary contributor to gorge formation. Whereas, The Peninsular rivers flow in comparatively shallow valleys. These are more or less completely graded valleys. The rivers have little erosional activity to perform.
- **Statement 3 is correct:** The **Brahmaputra** is one of the longest rivers of the world. Its drainage system is spread in three countries- China (Tibet), India and Bangladesh. It is known as the **Yarlung Tsangpo River in Tibet**, the Brahmaputra, Lohit, Siang, and Dihang in India, and the Jamuna in Bangladesh. The Brahmaputra has its **origin in the Chemayungdung glacier of the Kailash range near the Mansarovar lake.** Mariám La is a mountain pass to the southeast of Lake

Manasarovar. It separates the source of the Brahmaputra from the Manasarovar Lake.

Q.20) Ans: B

Exp:

- **Statement 1 is incorrect:** The **converging plate margins and mid-oceanic ridges are areas of most known volcanic activity and the earthquakes.** Nearly 70 percent of earthquakes occur in the Circum-Pacific belt. The Ring of Fire, also referred to as the Circum-Pacific Belt, is a path along the Pacific Ocean characterized by active volcanoes and frequent earthquakes. The majority of Earth's volcanoes and earthquakes take place along the Ring of Fire. Another 20 percent of earthquakes take place in the Mediterranean-Himalayan belt including Asia Minor, the Himalayas and parts of north-west China.
- **Statement 2 is correct:** **Tsunamis are caused by violent seafloor movement** associated with earthquakes, landslides, lava entering the sea, seamount collapse, or meteorite impact. Most tsunamis are generated by earthquakes. A disturbance that displaces a large water mass from its equilibrium position can cause a tsunami. Although relatively infrequent, violent volcanic eruptions also represent impulsive disturbances, which can displace a great volume of water and generate extremely destructive tsunami waves in the immediate source area.

Q.21) Ans: A

Exp:

- **Statement 1 is correct:** A **seamount is a submerged volcanic peak that hasn't ever reached the ocean's surface.** Whereas, **Guyots are seamounts that have been built above sea level.** Erosion by waves destroyed the top of the seamount resulting in a flattened shape. Due to the movement of the ocean floor away from oceanic ridges, the sea floor gradually sinks and the flattened guyots are submerged to become undersea flat-topped peaks.
- **Statement 2 is correct:** **Ocean trenches are long, narrow depressions on the seafloor.** These chasms are the **deepest parts of the ocean**—and some of the deepest natural spots on Earth. Ocean trenches are a **result of tectonic activity**, which

describes the movement of the Earth's lithosphere. In particular, ocean trenches are a feature of convergent plate boundaries, where two or more tectonic plates meet. At many convergent plate boundaries, dense lithosphere melts or slides beneath less-dense lithosphere in a process called subduction, creating a trench.

- **Statement 3 is incorrect:** The **Continental Shelf is the seaward extension of the continent** from the shoreline to the continental edge. Whereas, the continental slope connects the continental shelf and continental rise. Continental shelf ends where there is a sharp increase in the slope.

Q.22) Ans: D

Exp:

- **Statement 1 is correct:** Son Meta-Preference is the phenomena where parents continue to produce children until the desired number of sons are born. The various economic, religious, social and emotional desires and norms that favor males and make females less desirable. Often parents expect sons to provide financial and emotional care, especially in their old age, add to family wealth and property etc. All these lead to Gender inequality-the social process by which men and women are not treated equally.
- **Statement 2 & 3 is correct:** TFR is the average number of children expected to be born to a woman during her reproductive span of 15-49 years. **Sikkim recorded the lowest TFR**, with one woman bearing 1.1 children on average; **Bihar recorded the highest TFR** of three children per woman (National Family Health Survey-2019-20). **Kerala was the only state that experienced an increase in TFR to 1.8 from 1.6 in 2015-16 (NFHS-4)**. "Replacement level fertility" is the total fertility rate—the average number of children born per woman—at which a population exactly replaces itself from one generation to the next, without migration. According to the UN Population Division, a total fertility rate (TFR) of about **2.1 children per woman is called replacement-level fertility**.

Q.23) Ans: B

Exp:

- **Statement 1 is incorrect:** Weathering and erosion are processes by which rocks are broken down and

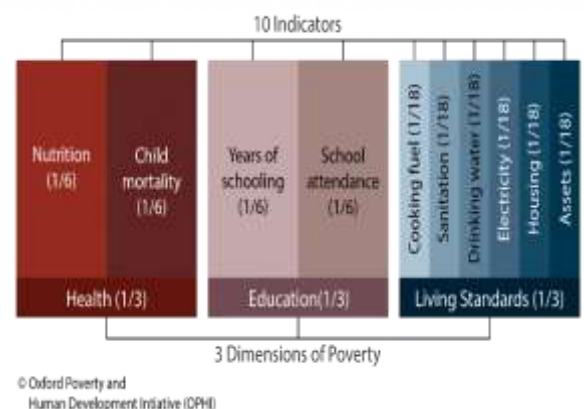
moved from their original location. They differ based on whether a rock's location is changed. The **weathering process degrades a rock without moving it** and the weathered materials are not displaced, whereas **erosion carries rocks and soil away from their original locations** involving the displacement of the eroded materials.

- **Statement 2 is correct:** When rocks and minerals begin to break down and dissolve, it's called weathering. Agents responsible for weathering include **ice, salts, water, wind and plants and animals**. Water is also a major agent of erosion on Earth. Rain, rivers, floods, lakes, and the ocean carry away bits of soil and sand and slowly wash away the sediment. Rainfall produces four types of soil erosion: splash erosion, sheet erosion, rill erosion, and gully erosion.
- **Statement 3 is correct:** Temperature changes contribute to mechanical weathering in a process called thermal stress. Changes in temperature cause rock to expand (with heat) and contract (with cold). As this happens over and over again, the structure of the rock weakens. High temperatures along with greater rainfall increase the rate of chemical weathering.

Q.24) Ans: D

Exp:

- **Pair 1 is correct:** The global Multidimensional Poverty Index (MPI) is an international measure of acute multidimensional poverty covering over 100 developing countries. It complements traditional monetary poverty measures by capturing the acute deprivations in health, education, and living standards that a person faces simultaneously. Multidimensional Poverty Index was launched by the UNDP and the OPHI in 2010.





- **Pair 2 is incorrect:** Global Gender Gap Report was first published in 2006 by the WEF. It benchmarks 156 countries on their progress towards gender parity in four dimensions: **Economic Participation and Opportunity, Educational Attainment, Health and Survival and Political Empowerment**. Over the Index, the highest possible score is **1 (equality) and the lowest possible score is 0 (inequality)**.
- **Pair 3 is incorrect:** The GII is an inequality index released by UNDP. It measures gender inequalities in three important aspects of human development—reproductive Health; Empowerment; and Economic status. The gender inequality index (GII) provides insights into gender disparities in health, empowerment and the labour market. **Unlike the human development index (HDI), however, higher values in the GII indicate worse achievements.**

Q.25) Ans: A

Exp:

- **Statement 1 is incorrect:** The distribution of the population according to different types of occupation is referred to as the occupational structure. **Primary activities include agriculture, animal husbandry, forestry, fishing, mining and quarrying** etc. Secondary activities include manufacturing industry, building and construction work etc. Tertiary activities include transport, communications, commerce, administration and other services.
- **Statement 2 is correct:** When two different air masses meet, the boundary zone between them is called a **front**. The process of formation of the fronts is known as **frontogenesis**. There are four types of fronts: (a) Cold; (b) Warm; (c) Stationary; (d) Occluded. In the northern hemisphere Frontogenesis happens in anti-clockwise direction and in the southern hemisphere, clockwise direction, due to the Coriolis effect. Mid-latitude cyclones or temperate cyclones or extra-tropical cyclones occur due to frontogenesis.
- **Statement 3 is correct:** The shape of bays and estuaries along a coastline can also magnify the intensity of tides. **Funnel-shaped bays greatly change tidal magnitudes**. When oceanic tidal bulges hit wide continental margins, the height of the tides can be magnified. Strong offshore winds can move water away from coastlines, exaggerating low tide exposures.

Q.26) Ans: D

Exp:

- **Statement 1 is correct:** Salinity is the term used to define the **total content of dissolved salts in seawater**. It is calculated as the **amount of salt (in gm) dissolved in 1,000 gm (1 kg) of seawater**. It is usually expressed as parts per thousand (o/oo) or ppt. The Factors affecting the Salinity of the Oceans are -
 - Evaporation and precipitation.
 - By the influx of freshwater from rivers, and freezing and thawing of ice.
 - Wind affects salinity by transferring water to other areas.
 - The ocean currents contribute to the salinity variations.
 - Salinity, temperature and density of water are interrelated. Hence, any change in the temperature or density influences the salinity of an area.
- **Statement 2 is correct:** The Red Sea is a seawater inlet of the Indian Ocean, lying between Africa and Asia. It is regarded as one of the most saline water bodies in the world. The salinity ranges between 36 and 38 ‰. It is much saltier than the Indian Ocean. The average salinity of the Indian Ocean is 35. The low salinity trend is observed in the Bay of Bengal due to influx of river water by the river Ganga.
- **Statement 3 is correct:** The relationship between salinity in the sea and the water cycle is well documented. Changes in salinity could also affect water currents and can tell us about changes in evaporation or rainfall in that particular location. Where more rain or snow falls over the ocean, it dilutes the salts in the seawater there. As a result, the water becomes fresher with time. If seawater becomes saltier, it may mean that rates of evaporation have increased or that precipitation has decreased over time. Changes in ocean currents can also influence salinity at the ocean's surface by moving salty water to places that are less salty, and vice versa.

REFERENCE: NCERT

Q.27) Ans: C

Exp:

- **Statement 1 is correct:** Khadar is the new alluvium and is deposited by floods annually,



which enriches the soil by depositing fine silts. It is found in the lower levels in the plains near the rivers. It is a Loamy and porous soil. It is comparatively more fertile than Bhangar as new layers are deposited year after year during monsoonal floods. Whereas, Bhangar represents a system of older alluvium, deposited away from the flood plains. It is Found higher up in the plains at river terraces away from rivers. It is Clayey and non-porous in nature. It is Less fertile than Khadar. Both Khadar and Bhangar soils contain calcareous concretions (Kankars).

- **Statement 2 is incorrect:** The laterite soils develop in areas with high temperature and high rainfall. These are the result of intense leaching due to tropical rains. With rain, lime and silica are leached away, and soils rich in iron oxide and aluminium compound are left behind. Humus content of the soil is removed fast by bacteria that thrives well in high temperatures. These soils are poor in organic matter, nitrogen, phosphate and calcium, while iron oxide and potash are in excess.
- **Statement 3 is correct:** Excessive irrigation with dry climatic conditions promotes capillary action, which results in the deposition of salt on the top layer of the soil. In such areas, especially in Punjab and Haryana, farmers are advised to add gypsum to solve the problem of salinity in the soil.

REFERENCE: NCERT

Q.28) Ans: d

Exp:

When two different air masses meet, the boundary zone between them is called a front. The process of formation of the fronts is known as frontogenesis.

There are four types of fronts:

(a) Cold; (b) Warm; (c) Stationary; (d) Occluded.

When the front remains stationary, it is called a **stationary front**.

Statement 1 is incorrect. When the cold air moves towards the warm air mass, its contact zone is called the **cold front**, whereas if the warm air mass moves towards the cold air mass, the contact zone is a **warm front**.

Statement 2 is incorrect. If an air mass is fully lifted above the land surface, it is called the **occluded front**.

Statement 3 is correct. The **fronts occur in middle latitudes** and are characterized by a steep gradient in temperature and pressure. They bring abrupt changes in temperature and cause the air to rise to form clouds and cause precipitation.

Q.29) Ans: c

Exp:

Options 2 and 3 are correct: Vog is a hazy mixture of **SO₂ gas and aerosols (tiny particles or droplets)** which are primarily **sulfuric acid and other sulfates (SO₄) compounds**.

Aerosols are **created when SO₂ and other volcanic gases combine** in the atmosphere and interact chemically with oxygen, moisture, dust, and sunlight over minutes to days.

As SO₂ is **released from the summit** and **east rift eruptive vents**, it reacts in the atmosphere with oxygen, sunlight, moisture, and other gases and particles and, within hours to days, converts to fine particles, which scatter sunlight, causing the visible haze.

SO₂ is a colorless, irritating gas that has an acrid odor like fireworks or a burning match. It is also emitted from sources such as fossil fuel power plants and motor vehicles

Fine particles consist of particulate matter less than 2.5 micrometers in diameter and are referred to as **PM_{2.5}**. These particles are smaller than the width of a human hair. **PM_{2.5} in vog is mainly composed of acid and neutral sulfate particles**. Other sources of PM_{2.5} include vehicle exhaust and smoke from fires.

Option 1 and 4 are not correct: Vog contains **mostly SO₂ and acid particles**, in contrast to urban, industrial, and other pollution sources, which also contain additional toxic contaminants, such as ozone, NO₂, and gaseous hydrocarbons.

Q.30) Ans: b

Exp:

Statement 1 is incorrect. Below the exosphere, molecules and atoms of atmospheric gases constantly collide with each other. However, the air in the exosphere is so thin that such collisions are very rare.



Gas atoms and molecules in the exosphere move along "**ballistic trajectories**", reminiscent of the arcing flight of a **thrown ball (or shot cannonball)**, as it gradually curves back towards Earth under the pull of gravity. Most gas particles in the exosphere zoom along curved paths without ever hitting another atom or molecule, eventually arching **back down into the lower atmosphere due to the pull of gravity**.

Statement 2 is correct. : At this distance, radiation pressure from sunlight exerts more force on hydrogen atoms than does the pull of Earth's gravity. A **faint glow of ultraviolet radiation scattered by hydrogen atoms** in the uppermost atmosphere has been detected at **heights of 100,000 km (62,000 miles)** by satellites. This **region of UV glow** is called the **geocorona**.

Statement 3 is correct. Although the exosphere is technically part of Earth's atmosphere, in many ways it is part of outer space. Many satellites, including the **International Space Station (ISS)**, **orbit within the exosphere or below**. The ISS loses about **2 km (1.2 miles)** in altitude each month to such "**orbital decay**", and must periodically be given an upward boost by rocket engines to keep it in orbit.

Q.31) Ans: b

Exp:

Option 1 is incorrect: Revolution of the earth around the Sun is not responsible for the variations in insolation at the surface of the Earth.

Variability of Insolation at the Surface of the Earth:

- The amount and the intensity of insolation vary during a day, in a season, and in a year.
- The factors that cause these variations in insolation are:
 - The **rotation of the earth** on its axis
 - The **angle of inclination** of the **sun's rays**
 - The **length of the day**
 - The **transparency** of the **atmosphere**
 - The **configuration of land** in terms of its aspect.

Hence the correct option is b.

Q.32) Ans: c

Exp:

Statement 1 is correct. Jet streams are currents of air high above the Earth. They **move eastward at altitudes of about 8 to 15 kilometers (5 to 9 miles)**. They **form where large temperature differences exist** in the atmosphere. The tropopause is one such spot. Jet streams are so fast and powerful that airplanes have difficulty flying against them. Pilots either fly with the jet stream or above it; they do not attempt to fly against it.

Statement 2 is correct. An air current is a flowing movement of air within a larger body of air. Air currents flow in the atmosphere, the layers of air surrounding the Earth. They **form because the sun heats the Earth unevenly**. As the sun beams down on the Earth, it **warms some areas**, particularly the **tropics, more than others, such as the poles**. As the Earth is heated, it warms the air just above it. The warmed air expands and becomes lighter than the surrounding air. It rises, creating a warm air current. **Cooler, heavier air then pushes in to replace the warm air, forming a cool air current**. Jet streams are air currents in the highest part of the atmosphere.

Q.33) Ans: a

Exp:

Statement 1 is correct. **Kelp forests are underwater areas** with a high density of kelp. They are recognized as one of the **most productive and dynamic ecosystems** on Earth. They occur worldwide throughout **temperate and polar coastal oceans**.

Statement 2 is incorrect. The **Photic Zone is a well-lit zone of oceans and highly productive**. This isn't necessarily a problem for oceans away from deserts, e.g. the Arabian Sea near Western ghats.

Statement 3 is incorrect. A recent discovery to play a significant role in **oceanic primary production** is the **micronutrient iron**. This is used as a cofactor in enzymes involved in processes such as nitrate reduction and nitrogen fixation. A **major source of iron to the oceans** is dust from the **Earth's deserts**, picked up and delivered by the wind as **aeolian dust**. In regions of the ocean that are distant from deserts or that are not reached by dust-carrying winds (for example, the Southern and North Pacific ocean),



the lack of iron can severely limit the amount of primary production that can occur.

Q.34) Ans: b

Exp:

Normally, **temperature decreases with an increase in elevation**. It is called a **normal lapse rate**. At times, the situation is reversed and the **normal lapse rate is inverted**. It is called **Inversion of temperature**.

Statement 1 is correct: Though inversion of temperature denotes local and temporary conditions of the atmosphere, there are several climatic effects of inversion which are of great significance to man and his economic activities. **Fog is formed** due to the situation of **warm air above and cold air below** because the warm air is cooled from below and resultant condensation causes the formation of tiny droplets around suspended dust particles and smokes during winter nights. The smoke coming out of houses and chimneys intensify fogs and become responsible for the occurrence of urban smogs. **Diffusion of dust, smoke, and other air pollutants is likewise limited due to calm winds**.

Statement 2 is correct: Temperature inversions affect air pollution because they change the dynamics of air movement. Warm air rises in the atmosphere because it's less dense and, therefore, more buoyant than the cooler air above it. However, a **temperature inversion prevents this vertical movement, also known as convection**. Stated differently, inversions function like an atmospheric lid or blanket. This **smothering effect traps air pollutants**, they get collected beneath the inversion layer and spread horizontally to fill the lower strata of the atmosphere. This can **increase their concentration** in the **lower-lying areas**.

Statement 3 is incorrect: Inversion of temperature **causes atmospheric stability** which stops upward (ascent) and downward (descent) movements of air. The atmospheric stability **discourages rainfall and favors dry conditions**.

Q.35) Ans: c

Exp:

Clouds are a **mass of minute water droplets or tiny crystals of ice** formed by the condensation of the water vapour in free air at a considerable elevation.

As the clouds are formed at some height over the surface of the earth, they take various shapes. According to their shape, density, opaqueness etc.

clouds are of four types viz.

cirrus, cumulus, stratus and nimbus.

Cirrus Clouds

These clouds are **formed at high altitudes (8,000 - 12,000m)**. They are thin and detached clouds having a feathery appearance.

Cumulus Clouds

Cumulus clouds **look like cotton wool**. They are generally **formed at middle height of 4,000- 7,000 m**.

Stratus Clouds

These clouds are **layered clouds** covering large portions of the sky. These clouds are generally **formed either due to loss of heat or the mixing of air masses** with different temperature.

Nimbus Clouds

Nimbus clouds are **black or dark gray**. They form at middle levels or very near to the surface of the earth. These are **extremely dense and opaque to the rays** of the sun. **Hence option 3 is correct.**

Q.36) Ans: c

Exp:

Zones of Ocean:

The ocean can be divided into many zones. The **ocean bottom is the benthic zone** and the **water itself (or the water column) is the pelagic zone**. The neritic zone is that part of the pelagic zone that extends from the high tide line to an ocean bottom less than 600 feet deep. **Water deeper than 600 feet is called the oceanic zone**, which itself is divided on the basis of water depth into the epipelagic, mesopelagic, and bathypelagic zones.

statement 1 is correct. The **epipelagic (euphotic) zone**, also called the **sunlit zone**, receives enough sunlight to support photosynthesis. The temperatures in this zone range anywhere from 40 to -3°C (104 to 27°F). The ocean bottom is the benthic zone.



statement 2 is incorrect. The **mesopelagic (disphotic) zone**, where **only small amounts of light penetrate**, lies below the epipelagic zone. This zone is often referred to as the Twilight Zone due to its scarce amount of light. **Temperatures** in the mesopelagic zone range from **5 to 4°C (41 to 39°F)**. The **pressure is higher here**; it can be up to 1,470 pounds per square inch (10,100,000 Pa) and increases with depth.

statement 3 is correct. **90% of the ocean lies in the bathypelagic (aphotic) zone** into which **no light penetrates**. This is also called the **midnight zone**. **Water pressure is very intense** and the **temperatures are near freezing** (range 0 to 6°C (32 to 43°F)).

Q.37) Ans: A

Exp:

- **Pair 1 is correct:** **Gabbro** is an example of **Plutonic rocks** that are **igneous rocks**, formed at some **depth in the earth's crust**. They have cooled and solidified slowly so that **large, easily-recognised crystals** have been able to form. E.g. **Granite** and **diorite** etc.
- **Pair 2 is correct:** **Sedimentary rocks** are formed from **sediment accumulated over long periods, usually under water**. They are distinguished from other rock types in their characteristic **layer formation** and are termed **stratified rocks**. **Sandstone** is an example of **Mechanically formed sedimentary rocks** that have been formed from the **accumulation of materials** derived from other rocks which have been **cemented together**.
- **Pair 3 is incorrect:** **Limestone** is an example of **Organically formed sedimentary rocks** that are formed from the **remains of living organisms** such as **corals or shellfish**, whose fleshy parts have been **decomposed**, leaving behind the **hard shells**. The most common rocks formed in this way are of the **calcareous type**. Another example is **chalk**.
- **Metamorphic rocks** are those rocks whose **original character and appearance** have been greatly **altered** by forces during intense earth movements. E.g. **slate, marble, quartzite** etc.

Reference: G C Leong

Q.38) Ans: B

Exp:

- **Composite Volcanoes:** These volcanoes are characterised by **eruptions of cooler and more viscous lavas** than **basalt**. These volcanoes often result in **explosive eruptions**. Along with lava, large quantities of **pyroclastic material and ashes** find their way to the ground. This material **accumulates** in the **vicinity** of the vent openings leading to **formation of layers**, and this makes the mounts appear as **composite volcanoes**.
- **Shield volcanoes:** The shield volcanoes are the **largest of all** the volcanoes on the earth. These volcanoes are **mostly made up of basalt**, therefore these volcanoes are **not steep**. They are generally characterised by **low-explosivity**.
- **Caldera:** These are the **most explosive** of the earth's volcanoes. They are usually so explosive that when they erupt, they tend to **collapse on themselves** rather than building any tall structure. The **collapsed depressions** are called **calderas**.
- **Mid-oceanic ridge volcanoes:** These volcanoes occur in the **oceanic areas**. There is a system of mid-ocean ridges **more than 70,000 km long** that stretches through all the ocean basins. The central portion of this ridge experiences frequent eruptions.

Reference: NCERT, Page no. 27

Q.39) Ans: C

Exp:

- **Statement 1 is incorrect:** **Fold mountains** are those mountains that are formed because of **large-scale earth movements**, where stresses are set up in the earth's crust. When such forces are initiated, rocks are subjected to **compressive forces** that produce wrinkling or folding along the **lines of weakness**. The great fold mountains of the world are the Himalayas, Rockies, Andes and Alps.
Block mountains: When earth's crust bends, folding occurs, but when it **cracks, faulting** takes place. Faulting may be caused by **tension or compression**, forces which **lengthen or shorten the earth's crust**, causing a **section** of it to **subside** or to **rise** above the surrounding level.
 - **Western Ghats** are **Block mountains** and the **faulted edge of the Deccan Plateau**. They are believed to have formed during the breakup of the



supercontinent around 60-70MYA (Cretaceous period).

- **Statement 2 is correct:** Volcanoes that build up from the material ejected from fissures in the earth's crust are called **volcanic mountains**. Volcanic mountains are often called **mountains of accumulation**. They are common in **Circum-Pacific belt** and include such volcanic peaks as **Mount Fuji (Japan), Mt. Mayon (Philippines), Mt. Merapi (Sumatra)** etc.
- **Statement 3 is correct:** Residual mountains are mountains evolved by **denudation**, where the general level of the land has been lowered by the **agents of denudation**. Residual mountains may also **evolve from plateaus** which have been **dissected by rivers** into **hills and valleys**. Examples of dissected plateau where the down-cutting streams have eroded the uplands into **mountains of denudation** are the Highlands of **Scotland, Scandinavia** and the **Deccan Plateau**.

Reference: G C Leong

Q.40) Ans: D

Exp:

- **Statement 1 is correct:** Mass movements transfer the mass of rock debris down the slopes under the direct **influence of gravity**. Rapid movements are mostly prevalent in **humid climatic regions** and occur over **gentle to steep slopes**. Movement of **water-saturated clayey or silty earth materials** down **low-angle terraces or hillsides** is known as **earthflow**.
- **Statement 2 is incorrect:** Slump is the **slipping** of one or several units of **rock debris** with a **backward rotation** with respect to the slope over which the movement takes place. **Rapid rolling or sliding** of earth debris **without backward rotation** of mass is known as a **debris slide**. Debris fall is **nearly a free fall** of earth debris from a vertical or overhanging face. **Sliding of individual rock masses** down bedding, joint or fault surfaces is a **rockslide**.
- **Statement 3 is correct:** In the **absence of vegetation cover** and with **heavy rainfall**, thick layers of weathered materials get **saturated with water** and either **slowly or rapidly flow** down along **definite channels**. It is called **Mudflow**. It looks like a **stream of mud** within a valley.

Reference: NCERT Page no. 53

Q.41) Ans: B

Exp:

River capture and cataracts are evidenced in the **upper or mountain courses** of a river. **Meanders** are evidenced in the **middle or valley course** and **Ox-bow lakes** are evidenced in the **lower or plain course** of a river.

- **River capture:** It is also known as **river piracy or river beheading**. Its development is dependent on different **rates of back-cutting** (headward erosion) into a divide.
- **Cataracts:** These are liable to occur at **any part of the river course**, but they are **most numerous** in the **mountain course**. Due to **unequal resistance of hard and soft rocks** traversed by a river, the outcrop of a band of hard rock may cause a **river to 'jump' or 'fall'** downstream and **Rapids** are formed. Similar falls of greater dimensions are referred to as **cataracts**. When rivers plunge down in a **sudden fall of some height**, they are called **waterfalls**.
- **Meander:** It is the winding course (**swinging in loops**) of a river due to **irregularities of the ground**.
- **Ox-bow lakes:** In the lower course of a river, a **meander** becomes very much more pronounced. Because of **rapid erosion of the outside bank**, the river becomes a **complete circle** which is cut after a period of time and an **ox-bow lake** is formed.

Reference: G C Leong

Q.42) Ans: C

Exp:

- **Statement 1 is incorrect:** The **Savannah Climate** is characterized by the alternate hot, rainy season and the cool and dry season. This region experiences an extreme diurnal range of temperatures and is dominated by the trade winds. Due to the large amount of hunting and killing of wild animals, this region is also known as the **Land of Big Games**. The vegetation of Savanna is characterized by tall grasses and short trees. The **Masai** are cattle pastoralists, an important tribe associated with the **Savannah region**. They are found mainly in



Kenya and Tanzania. They are known for their cattle called **zebu**, which are used only for the purpose of drinking their blood and milk. These cattles are never used as draught animals or for meat by the Masai.

- **Statement 2 is incorrect:** The **Hot Wet Equatorial Climate** is characterized by the uniformity of temperature throughout the year. They are marked by the absence of winters and the dominance of cloudiness, heavy precipitation resulting in the land and sea breezes. These regions witness a great diversity of vegetation dominated by the evergreen trees like the mahogany, ebony, rosewood, palm trees, epiphytes, etc. **Lalang** or the tall grasses are a feature of these areas. These grasses crop up in large numbers as soon as the tree is cut and can cause choking of the underlying crops.

Source: G C Leong

Q.43) Ans: C

Exp:

- The **strait of Bab al Mandab** is present between the Red Sea and the Gulf of Aden. It is surrounded by Djibouti, an African country and Yemen. Egypt, Sudan, Eritrea, and Djibouti are the countries of Africa that border the Red Sea.
- The **Strait of Gibraltar** is found between the Atlantic Ocean and the Mediterranean Sea. It is surrounded by Morocco and Spain. Morocco, Algeria, Tunisia, Libya, and Egypt are the African countries bordering the Mediterranean Sea.
- The **Strait of Malacca** lies between Malaysia and Indonesia. It connects the Indian Ocean with the Pacific Ocean through the South China Sea. Singapore is situated at the strategic location in the strait of Malacca.
- The **Strait of Bosphorus** connects the Black Sea with the sea of Marmara and is surrounded by Turkey.

Source: Atlas

Q.44) Ans: A

Exp:

- **Statement 1 is correct:** When air remains over a homogenous area for a longer period of time, it acquires the characteristics of the area and is referred to as the **air mass**. These air masses are classified according to the source region. When the two different air masses meet, the boundary area between them is referred to as the front and the process of formation of fronts is called frontogenesis. There are four types of fronts namely **stationary, cold, warm, and occluded fronts**.
 - The **stationary front** is the one which remains fixed at one position.
 - The **cold front** is formed with the movement of the cold air mass towards the warm air mass.
 - The **warm front** is formed when the warm air mass moves towards the cold air mass.
 - The **occluded front** is formed when the two interacting air masses are lifted completely above the ground.
- **Statement 2 is incorrect:** The point of convergence of North East and the South East Trade winds are called the **Inter Tropical Convergence Zone** or the **ITCZ**. The trade winds are the easterlies as they originate from the east. The air at the tropics rises due to higher insolation and high pressure. The winds from the low pressure area in the tropics converge at this location near the equator leading to the formation of a convective cell called the **Hadley cell**.

Source: ncert geo 12th physical

Q.45) Ans: C

Exp:

- **Statement 1 is correct:** **Ozone gas** is found at an altitude of around 10 to 50 km from the surface of the Earth. It is mostly found in the stratosphere and prevents harmful ultraviolet radiation from reaching the Earth. However, ozone present at the ground level acts as the pollutant. **Carbon dioxide** is an important gas present in the atmosphere of the Earth. It is

transparent to the incoming solar radiation while it prevents the terrestrial radiations from the earth to the atmosphere. This property of carbon dioxide is responsible for the greenhouse effect and in warming the Earth's atmosphere. The concentration of carbon dioxide is continuously increasing in the atmosphere and hence is responsible for global warming.

- **Statement 2 is incorrect:** The Earth's atmosphere is composed of numerous gases, water vapor, and dust particles. Nitrogen constitutes about 78% of the entire gases present in the atmosphere. Oxygen is almost negligible in quantity at a height of around 120 km while the carbon dioxide and water vapor are found only upto 90 km from the surface of the earth.
- **Statement 3 is correct:** The concentration of water vapor changes in the atmosphere with the change in the altitude. It is around 4% by volume in the warm and the tropical latitudes while its concentration decreases to less than 1% near the tropics. Its concentration also decreases towards the poles from the equator.

Source: ncert 12th geo pg 76

Q.46) Ans: D

Exp:

- A number of endogenic and exogenic processes are responsible for bringing about the changes in the surface of the Earth. These are known as the **Geomorphic processes** and they act with the help of Geomorphic agents like the running water, wind, ice, etc.
- **Diastrophism** is an endogenic process in which the force to bring about visible changes on the surface of the earth emanates from within the earth. In this, movement, building up, and elevation of the Earth's crust happens. Epeirogenic processes are a part of Diastrophism in which the uplift or warping of large portions of the Earth's crust occurs.
- **Erosion** causes the deposition and the transportation of materials or rocks which may or may not have been subjected to weathering. Erosion is a degradation process

and is controlled by the Kinetic energy of the acting agent which may be in the form of wind, running water, glacier, waves, and groundwater.

- **Salt crystallization** is the process of salt weathering which is favored in the areas of alternate dry and wet conditions. Larger temperature changes can cause expansion and contraction of the rocks and may lead to the formation of the exfoliation domes or sheets. Hence, the expansion and contraction of the existing structure due to temperature changes or the movement of the rocks causes exfoliation.

Source: Ncert 12th physical pg 51, 54

Q.47) Ans: D

Exp:

- **Statement 1 is correct:** The Bandicoot is a robotic machine that is engineered for cleaning any type of sewer manholes. The robot consists of two major units, a stand unit and a robotic drone unit. The drone unit which will dive into the manholes for the cleaning operations or unblocking operations. The aim is to eliminate manual scavenging, save humans from manhole accidents and adding dignity to human life.
- **Statement 2 is correct:** The Member of Parliament Local Area Development scheme funds can be used for developing model villages under the Saansad Adarsh Gram Yojana (SAGY). The MPs would be free to identify a suitable Gram Panchayat for developing Adarsh Gram, other than his/her own village or that of his/her spouse.
- **Statement 3 is correct:** The Ministry of Human Resource Development, the Ministry of Housing & Urban Affairs, and All India Council for Technical Education have jointly launched an online portal called **The Urban Learning Internship Program (TULIP).** The program aims to provide internship opportunities to fresh graduates in all Urban Local Bodies and Smart Cities across the country. It would help enhance the value-to-market of India's graduates and help create a potential talent pool in diverse fields.

Q.48) Ans: A

Exp:

- **Statement 1 is correct:** Atolls are basically circular oceanic reefs that surround a large central lagoon. It is the last stage of oceanic reefs. It starts with Fringing reefs then moves towards barrier reefs and later forms an atoll.
- **Statement 2 is correct:** Addu atoll, also called as Seenu atoll was in news as Indian Cabinet proposed to establish Indian Consulate. It is the southernmost atoll of Maldives. It has strategic importance in Indian Ocean and also is the largest city of the archipelago.
- **Statement 3 is incorrect:** Addu atoll lies to the south of equator and is a natural lagoon that has easy anchorage and accessibility through four natural channels. Safe harbour for sea vessels.

Reference: <https://www.thehindu.com/news/international/no-decision-yet-for-indian-consulate-in-addu-atoll-solih/article34772671.ece>

Q.49) Ans: B

Exp:

- The Horn of Africa consists of the internationally recognized countries namely **SEED (Somalia, Ethiopia, Eritrea, and Djibouti)**.
- In Ancient India and in medieval times it was also called the "Land of Barbaria and Ethiopians". The Horn of Africa is the easternmost peninsula of the African continent with its boundaries to the Red sea and extends to the Gulf of Aden.

Reference:

<https://www.thehindu.com/opinion/lead/Horn-of-Africa-why-India-should-care-more/article15596868.ece>

Q.50) Ans: A

Exp:

- **Statement 1 is correct:** Indian Ocean Dipole is an atmosphere-ocean coupled phenomenon of Indian Ocean. The prime cause of these phenomena are the difference in temperatures of Arabian Sea and Bay of Bengal.
- **Statement 2 is incorrect:** The early onset of the monsoon has no direct impact on the quantum of

rainfall. For example, in 2014 and 2015 the monsoon took 42 days and 22 days to cover the nation yet the rainfall quantum measures the same i.e.- deficient rainfall.

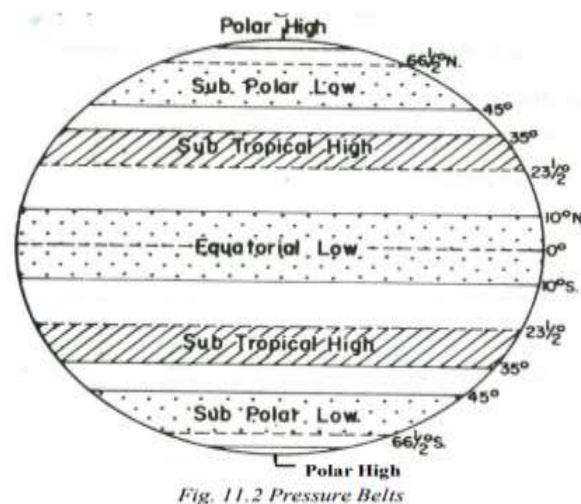
- **Statement 3 is incorrect:** A **positive IOD** is associated with cooler eastern Indian ocean sea surface temperature and warmer in western side. A **negative IOD** is associated with warmer eastern Indian Ocean sea surface temperature and cooler western margins.

Reference: <https://indianexpress.com/article/explain-ed/southwest-monsoon-rainfall-season-kerala-coast-7360834/#:~:text=The%20time%20of%20monsoon%20onset,or%20in%20the%20monsoon's%20progress.&text=Although%20it%20is%20too%20early,the%20normal%20of%20170%20mm.>

Q.51) Ans: C

Exp:

- **Equatorial low pressure belt** extends from the equator to 10°N and 10°S latitude. Due to excessive heating horizontal movement of air is absent here and only conventional currents are there. Therefore this belt is called **doldrums (the zone of calm)** due to the virtual absence of surface winds. Its Width may vary between 5°N and 5°S and 20°N and 20°S. The **position of the belt varies with the apparent movement of the Sun**. These are the **regions of convergence** because the winds flowing from sub-tropical high pressure belts converge here. This belt is also known as **Inter Tropical Convergence Zone (ITCZ)**.





- Near the equator the **sea level pressure is low** and the area is known as equatorial low. Along 30° N and 30° S are found the high-pressure areas known as the subtropical highs. Further pole wards along 60° N and 60° S, the low-pressure belts are termed as the sub polar lows. Near the poles the pressure is high and it is known as the polar high.

- **Statement 2 is correct-** The territories of India, Bhutan and China intersect in an area, called the Doklam Plateau, adjacent to the Chumbi Valley. It is a valley in the Himalayas that projects southwards from the Tibetan plateau, intervening between Sikkim and Bhutan. It is coextensive with the administrative unit Yadong County in the Tibet region of China. The Chumbi Valley is connected to Sikkim to the southwest via the mountain passes of Nathu La and Jelep La.

Q.52) Ans: D

Exp:

- **Statement 1 is Incorrect-** Gumati or Gomati is a river flowing through the north-eastern Indian state of Tripura and the district of Comilla in Bangladesh. A dam has been constructed near Dumbur on the river that has formed a lake covering 40 square kilometres.
- **Statement 2 is Incorrect-** The Bharathapuzha river rises in the **Eastern slopes of Anamalai hills of the Western Ghats** in Coimbatore district of **Tamil Nadu**. It enters the Palghat district of Kerala State through the Palghat gap. It finally discharges into the Arabian Sea near Ponnani town. **It is not a tributary of the Cauvery River.** It is also known as the Nila or Ponnani River, or Kuttippuram River, is a river in India in the state of Kerala.
- **Statement 3 is Incorrect:** The Dzukou valley is situated between Manipur and Nagaland, which is the crater base of an extinct volcano and the most famous Dzukou Lily (a flower) is found only in this region. This valley is well known for its natural environment, seasonal flowers and flora & fauna. It is situated at an altitude of 2452 m above sea level.

Q.53) Ans: C

Exp:

- **Statement 1 is correct-** Aksai Chin is important for China, as it connects two backward provinces of China, viz., Tibet and Xinjiang. Aksai Chin is a region administered by China as part of its Xinjiang and Tibet autonomous regions, and constituting the eastern portion of the larger Kashmir region which has been the subject of a dispute between India and China since the late 1950s.

Q.54) Ans: D

Exp:

Continental Drift Theory

- The theory of continental drift is most associated with the scientist Alfred Wegener. In the early 20th century, Wegener published a paper explaining his theory that the continental landmasses were “drifting” across the Earth, sometimes plowing through oceans and into each other. He called this movement continental drift.

Evidence in support of the Continental Drift theory

- **The Matching of Continents:** The most famous example cited in this regard is that of Eastern coast of South America and the Western coast of Africa fitting almost perfectly. This was later further proved using computer models.
- **Placer deposits:** Valuable placer deposits like gold can be found near the area of the Ghana coast in western Africa. while the source rocks are located in Brazil. Gold bearing veins are in Brazil. This led many to believe that the two continents lay side-by-side once upon a time.
- **Fossil distribution:** Irrespective of the massive marine barrier, many identical species of plants and animals can be found on two different continents. For example, Lemurs occur in India, Madagascar and Africa indicate that India was connected with Africa via Madagascar in the past.
- **Tillites:** These are sedimentary rocks formed out of deposits of glaciers. India's Gondwana rock system has its counterparts in at least six different land masses in the southern hemisphere. It is considered as an unambiguous evidence of climatic similarities as well as drifting of continents.



- **Rocks of the same age across continents:** A 2000 Million year old rock belt from Brazilian coast matches the one found in western Africa. The mountain systems of North America reappear in Scotland and Northern Ireland and match the age and structure of Appalachians of North America.

Source: Ncert class 11th chp 4

Q.55) Ans: D

Exp:

- **Pair 1 is incorrect :** The **Formosa strait or Taiwan strait** lies between China and Taiwan and joins the **South China sea and East China sea**. It lies in the Pacific ocean. It has been a contested water body due to tensions between Taiwan and China.
- **Pair 2 is incorrect:** The **strait of Tartary** lies between East Russia and Sakhalin Islands. It joins the **sea of Okhotsk and the sea of Japan**. It is located in the Pacific ocean. The sea of Okhotsk is one of the world's richest fishing grounds. The kuril islands in the sea of Okhotsk are contested between Japan and Russia.
- **Pair 3 correct:** The **strait of Otranto connects the Adriatic sea with the Ionian sea and separates Italy from Albania**. It has historical importance as the Romans used to carry troops eastwards through this strait. The main winds prevailing in the area are the Bora, a strong northeast wind that blows from the nearby mountains into the sea, and a southeasterly wind named the sirocco.
- **Pair 4 is correct :** The **Ten degree channel separates the Bay of Bengal from the Andaman sea**. It also **separates the Andaman islands from the Nicobar group of islands**. There are 4 major groups of Andaman islands called North Andaman, Middle Andaman, South Andaman and Little Andamans. The **Duncan passage separates little from the south Andaman**. Port Blair is the capital which lies in South Andaman.

Source: Atlas

Q.56) Ans: A

Exp:

- **Chagos archipelago-** It lies in the Indian ocean, south of the Maldivian islands. The island of Diego Garcia which serves as a military base for

the USA is located in this group of islands. It was recently in the news due to a tussle between the UK and Mauritius in the UN for control of the islands. The ICJ had asked the UK to hand over the islands to Mauritius as soon as possible. A judgement which the UK declined.

- **Christmas island** - It lies in the Indian ocean, NW to the coast of **Australia** and is administered by the same. It is closer to Asia than mainland Australia. It is known for its diverse fauna and coral reefs. It has a thick cover of monsoonal rainforests.
- **Gulf of Tonkin** - It lies in the South China sea, **Pacific ocean**. It is bordered by the countries of China and Vietnam. It is in the news due to chinese assertiveness increasing in the region over the surrounding waters.
- **Grand Banks-** It lies in the **Atlantic ocean**, on the East coast of Newfoundland, Canada. It is one of the richest fishing grounds of the world due to the mixing of cold Labrador and warm Gulf stream.

Q.57) Ans: D

Exp:

- **Pair 1 is correct - Popocatepetl** is one of Mexico's most active volcanoes. It became active after 50 years of dormancy in 1994 and has been erupting since then at irregular intervals. It is a stratovolcano/composite cone.
- **Pair 2 is incorrect - Anak Krakatau** is an Indonesian volcano which rises from the Sunda strait and separates the islands of Sumatra and Java. It means "child of Krakatau" because in 1883 the volcano of Krakatoa erupted so violently that two-thirds of it disappeared and from its caldera, emerged the Anak Krakatau.
- **Pair 3 is incorrect - Mauna Loa** is a volcano on the Hawaiian group of islands. It witnesses a hotspot type of eruption but has the shape of a shield volcano. It makes up more than 50 percent of Hawaii island. Eruptions are smooth and the lava is capable of travelling long distances due to low viscosity. It has not erupted since 1984.
- **Pair 4 is correct - Ojos del salados** is a volcano in Chile, South America. It is the world's highest active volcano. It is a stratovolcano and believed to have last erupted 1500 years ago. But it is considered active due to fumes coming out. It has



two summits of almost equal height, one in Chile and the other in Argentina.

Q.58) Ans: B

Exp:

- **Arctic Circle**, parallel, or line of latitude around the Earth, at approximately $66^{\circ}30' N$. Because of the Earth's inclination of about $23\frac{1}{2}^{\circ}$ to the vertical, it marks the southern limit of the area within which, for one day or more each year, the Sun does not set (about June 21) or rise (about December 21).
- The length of continuous day or night increases northward from one day on the Arctic Circle to six months at the North Pole. The countries part of this circle are:
 1. USA
 2. Canada
 3. Finland
 4. Denmark
 5. Norway
 6. Sweden
 7. Russia

Source: Ncert, Atlas

Q.59) Ans: A

Exp:

- **Pair 1 is incorrect:** Divergent/extensional/constructive boundaries are between two plates that are moving away from each other. Tensional stress is the dominant force and it produces features like Rift valleys and sea floor spreading. Examples of features are - East African rift valley, Mid Atlantic ridge.
- **Pair 2 is incorrect:** Convergent/ destructive plate boundaries are those where a plate with higher density dives under the other and is destroyed. They are of 3 types- Ocean-Ocean, Ocean-Continent, and continent-continent. Compressional forces are dominant and lead to features like- Trenching, Folding and Faulting of the plate.
- **Pair 3 is correct:** Parallel/transform/strike-slip boundary occurs when plates slide and grind against each other along a horizontal transform fault. In this kind of interaction, plates don't interact directly and hence not many features are associated except

occasional faulting and plates moving past each other.

Q.60) Ans: B

Exp:

- The **Somasila Hydro Power Project** is located in Andhra Pradesh. It is a dam built on the Pennar River. The Penna rises in the Nandi Hills in Chikkaballapur District of Karnataka state, and runs north and east through the states of Karnataka and Andhra Pradesh to empty into the Bay of Bengal. The river basin lies in the rain shadow region of Eastern Ghats.
- The **Chutak Hydro Power Project** is a "run of the river project", located in the Union Territory of Ladakh. It is located on the Suru River. The **Suru River** is a tributary of the Indus river that flows through the Kargil district of Ladakh. The river flows north into the Skardu District of Baltistan in Pakistan, where it joins the Indus at Nurla.
- The **Nimoo Bazgo Hydro Power Project** is a "run of the river" hydro power project, also located in the Union Territory of Ladakh. It is located on the Indus River. The **river Indus rises in Tibet, near Lake Mansarovar**. Flowing west, it enters India in the Ladakh district of Jammu and Kashmir. Several tributaries, the Zaskar, the Nubra, the Shyok and the Hunza, join it in the Kashmir region. The Satluj, the Beas, the Ravi, the Chenab and the Jhelum join together to enter the Indus near Mithankot in Pakistan. Beyond this, the Indus flows southwards eventually reaching the Arabian Sea, east of Karachi. . With a total length of **2900 km**, the Indus is one of the longest rivers. A little **over a third of the Indus basin is located in India** and the rest is in Pakistan.

Q.61) Ans: D

Exp:



- **Taiwan** is a country in East Asia. It shares maritime borders with the People's Republic of China (PRC) to the northwest, Japan to the northeast, and the Philippines to the south. The capital is Taipei.
- To the west is the Taiwan (Formosa) Strait, which separates Taiwan from the Chinese mainland.
- The East China Sea lies to its north, the Philippine Sea to its east, the Luzon Strait directly to its south and the South China Sea to its southwest.
- Taiwan lies on the Tropic of Cancer, and its general climate is marine tropical. Typhoons are most common in July, August and September.
- The warm Kuroshio (Japan Current), which flows northward to the east of Taiwan, moderates the island's temperatures.

Q.62) Ans: B

Exp:

- **Statement 1 correct and 2 is Incorrect: The Coral Triangle is a marine area located in the western Pacific Ocean**, which includes the waters of Indonesia, Malaysia, the Philippines, Papua New Guinea, Timor Leste and Solomon Islands.
 - The Coral Triangle is unique and important because it is home to the

highest concentration of marine species on the planet.

- The Coral Triangle, often called "the Nursery of the seas", is home to 600 corals or 76% of the world's known coral species.
- The Coral Triangle Day (held every June 9) is a celebration of the Coral Triangle, the world's epicentre of marine biodiversity.
- **Statement 3 is incorrect** - The CTI-CFF is a multilateral partnership formed by the governments of the six Coral Triangle countries (mentioned above) in 2009 to address the growing threats to the Coral Triangle. Under the CTI-CFF, the six countries signed a declaration to protect the Coral Triangle and committed to implement a **Regional Plan of Action (RPOA)** with five goals:
 - Designation of effectively managed seascapes
 - Application of an ecosystem approach to fisheries management;
 - Establishment of a fully functional marine protected area system
 - Strengthening climate change adaptation and resilience.
 - Improving the status of threatened marine species.

Q.63) Ans: A

Exp:

- **Statement 1 is correct:** The proportion of gases changes in the **higher layers of the atmosphere** in such a way that oxygen will be almost in negligible quantity at the height of 120 km. Similarly, carbon dioxide and water vapour are found only up to 90 km from the surface of the earth.

Table 8.1 : Permanent Gases of the Atmosphere

Constituent	Formula	Percentage by Volume
Nitrogen	N ₂	78.08
Oxygen	O ₂	20.95
Argon	Ar	0.93
Carbon dioxide	CO ₂	0.036
Neon	Ne	0.002
Helium	He	0.0005
Krypto	Kr	0.001
Xenon	Xe	0.00009
Hydrogen	H ₂	0.00005



- **Statement 2 is incorrect:** Nitrogen and oxygen make up nearly 99% of the clean, dry air. The remaining gases are mostly inert and constitute about 1% of the atmosphere.
 - Besides these gases, large quantities of water vapour and dust particles are also present in the atmosphere. These solid and liquid particles are of great climatic significance.

Q.64) Ans: C

Exp:

- Seabed 2030 is a collaborative project between the Nippon Foundation of Japan and the General Bathymetric Chart of the Oceans (GEBCO).
- It was launched at the **United Nations Ocean Conference in June 2017** and is aligned with the **UN's Sustainable Development Goal (SDG) 14** to conserve and sustainably use the oceans, seas and marine resources.
 - The United Nations Ocean Conference intended to be a game-changer in reversing the decline in the health of the ocean for people, planet and prosperity.
- **The project aims to bring together all available bathymetric data to produce the definitive map of the world ocean floor by 2030 and make it available to all.**
 - Bathymetry is the measurement of the shape and depth of the ocean floor.
 - In the past, satellites and planes carrying altimeter instruments have been able to provide large swathes of data about the ocean floor.
 - Despite many years of effort, less than 20 percent of the world ocean's seafloor has been mapped.
- Knowing the depth and shape of the seafloor (bathymetry) is fundamental for understanding ocean circulation, tides, tsunami forecasting, fishing resources, sediment transport, environmental change, underwater geo-hazards, infrastructure construction and maintenance, cable and pipeline routing and much more.

Source - <https://seabed2030.org/about-us>

Q.65) Ans: B

Exp:

- **The Ministry of Environment** recently in 2016 recategorized the industries **based on their pollution load** to create a clean and transparent working environment in the country and promote Ease of doing business.
- **White industries** (Industrial Sectors having Pollution Index score incl.& upto 20) - Practically non polluting industries are considered as white industries. For example, air coolers, air conditioning, biscuit factories etc. They don't need any green clearance.
- **Red Category (Industrial Sectors having Pollution Index score of 60 and above are under it)** - Cement, Petrochemical industry, pharmaceuticals, sugar, paper and pulp, nuclear plants, firecrackers, fertilizers, etc.
- **Orange category industries** (Industrial Sectors having Pollution Index score of 41 to 59) - Coal washers, glass manufacturing, stone crushers, aluminium and copper extraction from scrap industries.
- **Green category industries (Industrial Sectors having Pollution Index score of 21 to 40)** - Soap manufacturing, Tea processing etc.

Reference: <https://vikaspedia.in/energy/policy-support/environment-1/environment-ministry-releases-new-categorisation-of-industries>

Q.66) Ans: D

Exp:

- **Pair 1 is correct - Ropar lake:** It is a **MAN MADE freshwater riverine lake** and also wetland. It is located in **Punjab** and has an influx of river Sutlej.
- **Pair 2 is correct - Vembanad lake:** It is the longest lake in India as well as the largest lake in **Kerala** and is also recognized as a **Ramsar site**.
- **Pair 3 is Incorrect - Renuka lake:** Located in Sirmaur district of **Himachal Pradesh** and is also the largest lake in Himachal Pradesh. It has also been designated as a **Ramsar site** since 2005.
- **Pair 4 is Incorrect - Loktak lake:** Largest freshwater lake in Northeast India. It is located in **Manipur** and is also famous for **Phumdis**. The lake houses the only floating national park in the world, the Keibul Lamjao National Park, which is



the last refuge of the endangered brow-antlered deer or sangai, Manipur's state animal. Loktak Lake is one of the two montreux record sites in India, other is **Keoladeo National Park (Rajasthan)**.

Reference: 12th Indian Physical Geography NCERT

Q.67) Ans: D

Exp:

- **Endorheic basin:** It is a limited drainage basin that retains water and allows no outer flow of water to other bodies like oceans. But instead it converges into lakes and swamps.
 - **Pangong Tso lake:** It is long narrow, deep, endorheic (landlocked) lake situated at a height of more than 14,000 ft in the Ladakh Himalayas and is the world's highest saltwater lake. one-third of the Pangong Lake lies in India and the other two-thirds in China.
 - **Aral sea:** It is also an endorheic lake located **between Kazakhstan in the north and Uzbekistan** in the south. It has **shrunk** considerably since the 1960's. The shrinking of the Aral Sea has been called "one of the planet's worst environmental disasters".
 - **Caspian sea:** Its **World's largest inland body of water**. It lies **between Europe and Asia** and has an influx on many rivers like Volga, Ural, Terek etc.
 - **Sambhar lake:** **India's largest salt water lake** located in **Rajasthan**. It is also surrounded by **Aravali** mountains on all sides.

Reference:

https://en.wikipedia.org/wiki/Endorheic_basin

Q.68) Ans: C

Exp:

- **Statement 1 is correct:** A super cyclone is the one that has **maximum speeds of 120 knots** and **higher than normal temperatures**. The recent **cyclone Amphan** was classified as a super cyclone. It made its landfall in **West Bengal Sagar Islands**. **It is the second super cyclone to have formed over the Bay of Bengal in two decades.**

- **Statement 2 is incorrect:** Cyclone Taukate was not clubbed as a super cyclone. The recent two super cyclones are **Amphan of 2020 and another in 1999 at Paradip port**.
- **Statement 3 is incorrect:** Cyclone Taukate has **made a landfall in Gujarat** and has swept states like Kerala, Karnataka, Goa and Maharashtra. Whereas Hatiya islands of Bangladesh is on the eastern side. Similarly, Sagar island is situated in Bay of Bengal.
- **Statement 4 is correct:** The cyclone Taukate was named by **Myanmar, which translates to "gecko"** a highly vocal lizard in Burma.

Reference:

<https://www.thehindu.com/news/national/cyclone-taukate-on-the-wane-new-storm-likely-in-bay-of-bengal/article34598732.ece>

Q.69) Ans: B

Exp:

- **Pair 1 is incorrect:** **Barnadi wildlife sanctuary** is located in northern **Assam's Baksa and Udalguri** districts bordering Bhutan. The Sanctuary is bordered by the **Barnadi river** and **Nalapara river** to the west and east respectively. About 60% of the BWS is reported to be **grassland**, most of it is now grassy woodland. The main Forest types are **Tropical Moist Deciduous** along the northern edge of the Sanctuary, and **mixed scrub and grassland** in the southern part with few trees.
- **Pair 2 is correct:** **Dehing-Patkai elephant reserve** is located in **Upper Assam** (Dibrugarh and Tinsukia districts). **Dehing** is the **name of the river** that flows through this forest and **Patkai** is the **hill** at the foot of which the sanctuary lies. It is the **only sanctuary in India** which is home to **seven different species of wild cats** - tiger, leopard, clouded leopard, leopard cat, golden cat, jungle cat and marbled cat. It has the **highest concentration** of the rare endangered **White Winged Wood Duck**.
- **Pair 3 is incorrect:** **Pakhui Tiger Reserve**, also known as **Pakke Tiger Reserve**, is a Project Tiger reserve. It is located in the **Pakke Kessang district of Arunachal Pradesh**. It falls within the **Eastern Himalaya Biodiversity Hotspot**. This Tiger Reserve has won **India Biodiversity Award 2016** in the category of 'Conservation of threatened species' for its **Hornbill Nest Adoption Programme**.



Reference: OnlyIAS Magazine

Q.70) Ans: C

Exp:

- **Statement 1 is correct:** Gharial is a species of Asian crocodilian distinguished by their long, thin snouts which resembles a pot. They prefer to live in riverine habitats with deep, clear, fast-moving water and steep, sandy banks. It was once found across Bangladesh, Bhutan, India, Myanmar, Nepal and Pakistan. However, currently it survives in several severely fragmented populations in India and Nepal.
- **Statement 2 is incorrect:** They are categorized as critically endangered in IUCN red list, scheduled in Schedule I of Wildlife Protection Act, 1972 and Appendix I of CITES.
- **Statement 3 is correct:** In India, Gharials are present in Son River, Girwa River, the Ganges, Mahanadi River and the Chambal River. Satkosia gorge of Mahanadi river is the southernmost limit of its home range and the last for gharials in Odisha, which is the only state having all three species of crocodiles - gharial, mugger and saltwater crocodile.

Reference:

OnlyIAS Magazine

<https://www.newindianexpress.com/states/odisha/2019/jul/21/crocodiles-released-in-odishas-satkosia-gorge-for-pilot-study-2007119.html>

Q.71) Ans: C

Exp:

About Artemis Accord:

- The Artemis Accords, a blueprint for space cooperation and supporting the U.S. space agency's plans to return humans to the moon by 2024 and to launch a historic human mission to Mars.
- This accord ensures that space exploration is conducted in a safe, sustainable and transparent manner and in full compliance with international law.
- As more countries establish a presence in outer space, via research stations, satellites, or even rocket launches, these accords provide a set of principles to create a safe and transparent

environment that inspires exploration, science, and commercial activities.

- The other signatories to the accords are the U.S., Australia, Britain, Canada, Italy, Japan, Luxembourg, South Korea, the United Arab Emirates, and Ukraine. Brazil also said it plans to sign.
- Recently New Zealand became the eleventh signatory to the Artemis Accords.

Reference: OnlyIAS Magazine

Q.72) Ans: D

Exp:

About Pradhan Mantri Fasal Bima Yojana:

Launched in 2016, the flagship PMFBY insures farm losses against adverse weather events. It is a central scheme implemented by state agriculture departments as per central guidelines. Recently, the Central government modified certain provisions of the scheme:

- **Option A is correct:** Central Subsidy under PMFBY is limited for premium rates upto 30% for unirrigated areas/crops and 25% for irrigated areas/crops. Districts having 50% or more irrigated area will be considered as irrigated area/district.
- **Option B is correct:** States not to be allowed to implement the Scheme in subsequent Seasons in case of considerable delay by States in release of requisite Premium Subsidy to concerned Insurance Companies beyond a prescribed time limit. Cut-off dates for invoking this provision for Kharif and Rabi seasons will be 31st March and 30th September of successive years respectively.
- **Option C is correct:** Enrolment under the Scheme to be made voluntary for all farmers.
- **Option D is incorrect:** Provisioning of at least 3% of the total allocation for the Scheme to be made by Government of India and Implementing State Governments for administrative expenses. This shall be subject to an upper cap fixed by the Department of Agriculture Cooperation and Farmers Welfare (DAC&FW) for each State.

Reference:

<https://pib.gov.in/PressReleasePage.aspx?PRID=1603638>



Q.73) Ans: C

Exp:

About World Investment Report, 2021:

- The report is **released** by the **United Nations Conference on Trade and Development**.
- The World Investment Report focuses on **trends in Foreign Direct Investment (FDI)** worldwide, at the **regional and country levels** and **emerging measures** to improve its contribution to development.
- The report says that **India** is the **fifth largest recipient (\$64 billion in 2020)** of Foreign Direct Investment (FDI) in the world.
- The report has further said that the **global FDI flows** have been severely **hit by the pandemic** and they **plunged by 35% in 2020 to USD 1 trillion**. **COVID-19 led lockdowns** around the world slowed down existing investment projects and prospects of a recession led multinational enterprises (MNEs) to reassess new projects.

Reference: OnlyIAS Magazine

Q.74) Ans: B

Exp:

- **Statement 1 is correct:** The Model Tenancy Act is **applicable prospectively** and will not affect the existing tenancies. **Written agreement is a must** for all new tenancies. The agreement will have to be **submitted to the concerned district 'Rent Authority'**.
- **Statement 2 is incorrect:** A **landowner** or property manager may enter a premise in accordance with **written notice or notice through electronic medium** served to the tenant **at least twenty-four hours** before the time of entry.
- **Statement 3 is correct:** The **security deposit** to be paid by the tenant **should not exceed two months' rent for residential property (six months' rent in case of non-residential property)**, and should be a **minimum of one month's rent for non-residential property**. Under the Act, the landowner would give a **notice in writing three months before revising rent**. The landlord cannot hike the rent in the middle of the tenure.

Extra Edge by OnlyIAS:

Other key provisions of the Model Act:

- The law also speaks about **roles and responsibilities** of landlords and tenants.
- No landlord or property manager can **withhold any essential supply** to the premises occupied by the tenant.
- If **tenancy has not been renewed**, the tenancy shall be **deemed to be renewed on a month-to-month basis** on the same terms and conditions as were in the **expired tenancy agreement**, for a **maximum period of six months**.
- On the **expiry of an extended period of six months** of agreed tenancy period or the **termination of tenancy** by order or notice, the tenant shall be a tenant in default and liable to pay compensation of **double of the monthly rent for two months** and **four times** of the monthly rent **thereafter**.

Reference:

<https://pib.gov.in/PressReleasePage.aspx?PRID=1723636>

Q.75) Ans: C

Exp:

- **Statement 1 is incorrect:** The **SDG India Index** was launched in **2018** by NITI Aayog. It was developed in **collaboration with the United Nations**. The United Nations adopted Sustainable Development Goals in **2015**. The index determines **India's progress towards achieving the Sustainable Development Goals (SDGs) by 2030**.
- **Statement 2 is incorrect:** The **2018 edition** of SDG India Index covered **13 SDGs with 62 indicators**. The **2020 edition** covers **16 SDGs on 115 quantitative indicators**.
- **Statement 3 is correct:** States/UTs are classified based on the SDG India Index Score as follows:

- **Aspirant:** 0–49

Top 5 positions		Bottom 5 positions	
State	Score	State	Score
Kerala	75	Chhattisgarh,	61
Himachal Pradesh,	74	Nagaland, Odisha	
Tamil Nadu		Arunachal Pradesh,	60
Andhra Pradesh, Goa,	72	Meghalaya, Rajasthan,	
Karnataka, Uttarakhand		Uttar Pradesh	
Sikkim	71	Assam	57
Maharashtra	70	Jharkhand	56
		Bihar	52

Source: NITI Aayog

- **Performer:** 50–64
- **Front Runner:** 65–99
- **Achiever:** 100
- Currently, there are **no states** in the **aspirant** and **achiever** category. Around **15 states/UTs** are in the **performer** category and **22 states/UTs** in the **front runner** category. **Kerala** has **topped** the index with a score of 75. It was followed by **Himachal Pradesh** and **Tamil Nadu** with a score of 74. **Bihar, Jharkhand** and **Assam** were the **worst-performing states** in the SDG India index.

Reference

<https://indianexpress.com/article/explained/niti-aayog-sustainable-development-goals-ranks-explained-7343640/>

Q.76) Ans: A

Exp:

- **Option A is incorrect:** PM CARES will contribute through a **specifically designed scheme** to create a **corpus of Rs 10 lakh for each child** when he or she reaches 18 years of age. This corpus:
 - Will be used to give a **monthly financial support/ stipend** from **18 years of age**, for the **next five years** to take care of his or her **personal requirements** during the period of **higher education** and
 - On reaching the **age of 23 years**, he or she will get the corpus amount as **one lump-sum for personal and professional use**.
- **Option B is correct:** The child will be **assisted** in **obtaining an education loan** for **Professional courses / Higher Education in India** as per the existing Education Loan norms. The **interest on this loan** will be paid by the PM CARES. As an alternative, **scholarships equivalent** to the tuition fees / course fees for undergraduate/ vocational

courses as per Government norms will be provided to such children under Central or State Government Schemes. For children who are **not eligible** under the existing scholarship schemes, PM CARES will provide an **equivalent scholarship**.

- **Option C is correct:** All children will be enrolled as a **beneficiary under Ayushman Bharat Scheme (PM-JAY)** with a health insurance cover of Rs. 5 lakhs. The **premium amount** for these children **till the age of 18 years** will be paid by PM CARES.
- **Option D is correct:** PM CARES will also pay for expenditure on uniforms, textbooks and notebooks till 18 years of age.

Extra Edge by OnlyIAS

Other features of the scheme:

- **School Education: For children under 10 years:** The child will be given admission in the **nearest Kendriya Vidyalaya** or in a **private school** as a day scholar.
- **School Education: for children between 11-18 years:** The child will be given admission in **any Central Government residential school** such as **Sainik School, Navodaya Vidyalaya** etc.
 - In case the child is to be **continued** under the care of **Guardian/ grandparents/ extended family**, then he or she will be given admission in the **nearest Kendriya Vidyalaya** or in a **private school** as a day scholar.
- If the child is admitted in a **private school**, the **fees as per the RTE norms** will be given from the PM CARES.

Reference:

<https://pib.gov.in/PressReleasePage.aspx?PRID=1722719>

Q.77) Ans: B

Exp:

Recently **NASA** announced **two new Venus Missions—DaVinci+ and Veritas**. These will examine the **planet's atmosphere and geological features**.

- **DaVinci Plus:** **Davinci+** stands for **Deep Atmosphere Venus Investigation of Noble gases, Chemistry, and Imaging**. The aim of the mission is to measure the **planet's atmosphere** to gain insight into how it formed and evolved. It will also



aim to determine whether **Venus** ever had an **ocean** and was possibly habitable.

- **Veritas:** VERITAS stands for **Venus Emissivity, Radio Science, InSAR, Topography, and Spectroscopy**. The mission will map the planet's surface to understand its **geologic history** and investigate how it developed so differently than Earth.
- **New Horizon:** It was launched in **2006** by NASA to help us understand the edge of our solar system by making the first reconnaissance of the **dwarf planet Pluto** and venturing deeper into the **Kuiper Belt**.
- **Dawn Mission:** Dawn is a mission of NASA to the two most massive bodies in the main asteroid belt – **Vesta** and **Ceres**. It is part of the **Discovery programme** and was launched in **2007**.

Reference:

https://www.nasa.gov/mission_pages/newhorizons/overview/index.html

https://www.nasa.gov/mission_pages/dawn/mission/index.html

OnlyIAS

Q.78) Ans: B

Exp:

- **Statement 1 is incorrect:** The **National Institute of Rural Development and Panchayati Raj** is an autonomous organization under the **Ministry of Rural Development**. It is recognized as the Centre of Excellence of the **UN-ESCAP (Economic and Social Commission for Asia and Pacific)**. It is located in Hyderabad and aims at building capabilities for the functionaries of rural development, elected representatives of the Panchayati Raj Institutions, Bankers, NGOs, and other related functionaries with the aid of training, research, and consultancy.
- **Statement 2 is correct:** **Article 243G** of the Constitution of India deals with the powers, authority and the responsibilities of the Panchayats. The Panchayat Raj Institutions deals with the local self-governing institutions and was constitutionalized through the **73rd Constitutional Amendment Act, 1992**. It contains provisions regarding the Gram Sabha and the three-tier structure of the Panchayati Raj institutions. The **73rd Constitutional Amendment Act, 1992** added

a new Part IX to the Constitution which comprises Article 243 to 243-O. The act was also responsible for the addition of the Eleventh Schedule that contains 29 items that are associated with the Panchayats. It is related to Article 243-G.

Source: <http://nirdpr.org.in/overview.aspx> ;
Laxmikanth

Q.79) Ans: D

Exp:

- The **Comprehensive Test Ban Treaty** was made open for signature in 1996 and it bans nuclear testing in all environments including outer space and water. CTBT has been enforced yet. India initially participated in the negotiations of CTBT but did not sign it.
- **Partial Test Ban Treaty, 1963** aims to prohibit the nuclear test in the outer atmosphere, in the outer space, under the water, and in the State's territories that would have nuclear fallout in the territories of the other states. India is a member of the PTBT.
- **Treaty on the Non-Proliferation of nuclear weapons or the NPT, 1970** aims to prevent the spread of Nuclear weapons and also seeks to make use of the nuclear weapons for peaceful use. India is not a party to this treaty.
- The **Treaty on the Prohibition of nuclear weapons or the TNPW** includes the legally binding commitments and was enforced recently in 2020. It seeks to move towards total elimination of nuclear weapons and prohibit development, use, stockpiling, threatening, acquiring, testing of the nuclear weapons. The TPNW is the first treaty to establish a comprehensive ban on nuclear weapons, including their development, deployment, possession, use and threat of use.

Source: https://sipri.org/sites/default/files/2021-06/sipri_yb21_summary_en_v2_0.pdf; <https://www.nti.org/learn/countries/india/>

Q.80) Ans: A

Exp:

- **Statement 1 is incorrect:** The **Reform Based and Result Linked Revamped Distribution Scheme**



was launched to improve the efficiencies of the DISCOMs. It does not include the private sector entities. It aims to reduce the AT & C - Aggregate Technical and Commercial Losses to pan-India levels by 12-15% by 2024-25.

- **Statement 2 is incorrect:** A key feature of the Scheme is the use of prepaid Smart metering to be implemented in Public-Private-Partnership (PPP) mode. While in all 25 crore Smart meters are planned to be installed during the Scheme period, priority would be given to install prepaid Smart Meters in a mission mode in the first phase in
 - all Electricity Divisions of 500 AMRUT cities, with AT&C Losses > 15%
 - all Union Territories
 - MSMEs and all other Industrial and Commercial consumers
 - all Government offices at Block level and above
 - other areas with high losses.

However, due to the scattered nature of the agricultural connections, the agricultural consumers would be connected through the Feeder meters.

- **Statement 3 is correct:** The Scheme provides for annual appraisal of the DISCOM performance against predefined parameters including AT&C losses, ACS-ARR gaps, infrastructure upgrade performance, consumer services, hours of supply, corporate governance, etc. Hence, it seeks to reduce the Average cost of supply- Average revenue realized gap.

Source:

<https://pib.gov.in/PressReleasePage.aspx?PRID=1731473>

Q.81) Ans: D

Exp:

- AgriStack is a collection of technologies and digital databases that focuses on farmers and the agricultural sector. It will create a unified platform for farmers to provide them end to end services across the agriculture food value chain. AgriStack aims to create a database regarding the farmer, farms, and the crops.

- Recently, An MoU was signed by the Department of Agriculture, Cooperation and Farmers Welfare with Microsoft for the creation of a **Unified Farmer Service Interface** in its cloud computing services in a pilot project of 100 villages in the states of Uttar Pradesh, Haryana, Rajasthan, Madhya Pradesh, Gujarat, Andhra Pradesh. It marked the beginning of **AgriStack**.
- In order to create Agristack, the government is in the process of finalising “**India Digital Ecosystem of Agriculture** (IDEA)” which will lay down a framework for Agristack.

Source:

<https://www.downtoearth.org.in/news/agriculture/agristack-the-new-digital-push-in-agriculture-raises-serious-concerns-77613>

Q.82) Ans: B

Exp:

- **Statement 1 is incorrect:** **Hallmarking** is the certificate of purity that was provided by the **Bureau of Indian Standards** for the hallmarking of the gold and silver jewelry. So far hallmarking was voluntary. However, it was made mandatory from 16 June 2021 in a phased manner in selected districts where the jewelers were allowed to sell only 14, 18, and 22 carats of Gold. Hallmarking would protect the consumers and help them make informed choices for the purer gold. It would also enhance the credibility of the jewellery. The Jewellers with the annual turnover upto 40 Lakhs were exempted from the mandatory hallmarking.
- **Statement 2 is incorrect:** Although currently BIS or the Bureau of Indian Standards is the only national body that develops standards in the country. Yet, a number of PSUs and institutions can also frame standards for the products in their own domain. For example, the Food Safety and Standards Authority of India or FSSAI frames standards for the food items in the country.
- **Statement 3 is correct:** Annex 3 of the **Agreement on Technical Barriers to Trade** of the **WTO** contains provisions regarding the Code of Good Practice for the Preparation, Adoption and Application of Standards. India is

a signatory to this and hence Central, State, Local Governments or any Non-governmental organization that deals with Standardization has to ensure compliance to these practices.

Source:

<https://indianexpress.com/article/explained/what-is-gold-hallmarking-list-of-districts-where-it-is-mandatory-7361818/> ;

<https://pib.gov.in/PressReleaseFramePage.aspx?PRID=1723360>

Q.83) Ans: D

Exp:

- **Statement 1 is incorrect:** Cotton is a **Kharif crop** that is grown in the black soil areas with the temperature around 20-28 degrees Celsius and the rainfall between 55- 110 cm. It requires a minimum of 180 frost free days. Gujarat, Maharashtra, and Madhya Pradesh account for around 54% of the total cotton production of the country.
- **Statement 2 is incorrect:** **Minimum Support Prices** on cotton in India is fixed by the Government of India based on the recommendation of the **Commission on Agriculture Costs and Prices** or the CACP. Since 2008-09, it has been decided on two staple groups of Medium and Long. The cotton is divided into these groups based on the staple length.

Source:

ncert;

https://www.business-standard.com/article/international/half-of-global-cotton-growing-regions-may-face-severe-climate-risks-study-121062301854_1.html ;
https://cotcorp.org.in/marketing_faq.aspx#ans18

Q.84) Ans: B

Exp:

- **Gross Environmental Product** refers to the total value of goods and services that are provided by the ecosystem to the human beings annually and can be measured in monetary values. It determines the health of an ecosystem by measuring the amount of forest cover, the rivers, lakes, biodiversity, etc. in an ecosystem. Apart from measuring the

Gross Domestic Product or the GDP to ascertain the economic growth, it is also essential to measure the Gross Environment Product by measuring the services that an ecosystem provides to an economy. Recently, Uttarakhand became the first state in the country to take into account the GEP along with the GDP.

- While **GDP** would solely comprise mining, construction, transportation, Information and communication, Financial services, public administration, etc. The **GEP** would constitute Flood prevention, **mental health**, carbon sequestration, **sandstorm prevention**, **pollination**, Air purification, water purification, etc. Certain services carry the elements of **both GEP and GDP**. These are Agriculture, **Forestry**, Fishing, **Hydropower**, Recreation and Tourism, etc.

GDP and GEP



Source:

https://seea.un.org/sites/seea.un.org/files/1._gross_ecosystem_product_and_ecological_assets_0.pdf (page 12)

<https://www.downtoearth.org.in/blog/environment/introducing-gross-environment-product-in-uttarakhand-a-jargon-or-reality--78088>



Q.85) Ans: A

Exp:

- **Statement 1 is correct:** The **Great Barrier Reef**, a complex of coral reefs, shoals, and islets in the Pacific Ocean off the northeastern coast of Australia that is the longest and largest reef complex in the world. A large part of the reef is protected by the Great Barrier Reef Marine Park, which helps to limit the impact of human use, such as fishing and tourism. It was designated as a World Heritage site by UNESCO.
- **Statement 2 is incorrect:** Recently **UNESCO recommended the World Heritage Committee to add the Great Barrier Reef in the "Danger list" but Australia managed to defer the attempt.**
- **Statement 3 is correct:** The coral ecosystem has suffered mainly by bleaching acts which are the direct result of increased frequency of heat waves. Australian government has brought out The Reef 2050 Long Term Sustainability Plan for protecting and managing the Great Barrier Reef by 2050.
 - Other environmental pressures on the reef and its ecosystem include runoff, climate change accompanied by mass coral bleaching, dumping of dredging sludge and cyclic population outbreaks of the crown-of-thorns starfish.

Coral Bleaching

- Coral bleaching happens when corals lose their vibrant colors and turn white. But there's a lot more to it than that.
- Coral are bright and colorful because of microscopic algae called zooxanthellae. The zooxanthellae live within the coral in a mutually beneficial relationship, each helping the other survive.
- When the ocean environment changes—if it gets too hot, for instance—the coral stresses out and expels the algae. As the algae leaves, the coral fades until it looks like it's been bleached. If the temperature stays high, the coral won't let the algae back, and the coral will die.
- The leading cause of coral bleaching is climate change. A warming planet means a warming ocean, and a change in water temperature—

as little as 2 degrees Fahrenheit—can cause coral to drive out algae. Coral may bleach for other reasons, like extremely low tides, pollution, or too much sunlight.

Reference:

<https://www.thehindu.com/news/international/great-barrier-reef-should-be-listed-as-in-danger-un-committee-recommends/article34897826.ece>

Q.86) Ans: B

Exp:

- **Statement 1 is Incorrect:** India's IMR stands at 32 per 1,000 live births. Madhya Pradesh continues to have the worst IMR in the country at 48 while Nagaland has the best at 4, according to data released by the Registrar General of India. The largest decline in IMR has been registered by Mizoram with a decline from 15 to 5. Chhattisgarh has the highest death rate in the country at 8, while Delhi has the lowest at 3.
- **Statement 2 is correct:** Among large states, Kerala is the best performer being the only state with an IMR in single digits at 7. It is also the only state which has now achieved the United Nations' sustainable development goal in this regard.
- **Statement 3 is correct:** The Causes of Infant Mortality are mostly Preterm birth, low birth weight, Maternal pregnancy complications, Sudden infant death syndrome, Injuries (e.g., suffocation). etc. High infant mortality rates are generally indicative of unmet human health needs in sanitation, medical care, nutrition, and education because it is particularly sensitive to general structural factors, like socio-economic development and basic living conditions.

REFERENCE:

<https://thewire.in/health/india-infant-mortality-rate>

Q.87) Ans: A

Exp:



- **Statement 1 is correct:** The Special Economic Zones (SEZs) policy was launched in April, 2000. The Special Economic Zones Act, 2005, was passed by Parliament in May, 2005. Some incentives for SEZ include:

- Duty free import/domestic procurement of goods for development, operation and maintenance of SEZ units
- 100% Income Tax exemption on export income for SEZ units under Section 10AA of the Income Tax Act for first 5 years, 50% for next 5 years thereafter and 50% of the ploughed back export profit for next 5 years. (**Sunset Clause for Units will become effective from 01.04.2020**)
- Exemption from Central Sales Tax, Exemption from Service Tax and Exemption from State sales tax. These have now subsumed into GST and supplies to SEZs are zero rated under IGST Act, 2017.
- Other levies as imposed by the respective State Governments.
- Single window clearance for Central and State level approvals.

- **Statement 2 is correct:** As of 31st January 2021; 265 SEZs are operational in the country. Tamil Nadu has the highest number of operational SEZs (48), followed by Maharashtra (37).

- **Statement 3 is incorrect:** Mostly all labour laws are applicable in SEZs. As SEZs are to adhere to government regulations to protect worker interests, the major laws still apply, such as: Payment of workmen act, 1923. Payment of wages act, 1936 and minimum wages act, 1948. Payment of Bonus act, 1965.

and the **Andamans and Nicobar Islands**. It is also found across Southeast Asia and northern Australia. Its protection status is:

- IUCN Red List: **Least Concern**
- Wild Life (Protection) Act, 1972: **Schedule I**
- CITES: **Appendix I** (except the populations of Australia, Indonesia and Papua New Guinea, which are included in Appendix II).

- **Pair 2 is correct:** **Olive Ridley turtles** are the **smallest and most abundant** of all sea turtles found in the world. They are best known for their **unique mass nesting** called **Arribada**, where thousands of females come together on the same beach to lay eggs. They are found in **warm waters of the Pacific, Atlantic and Indian oceans**. The **Odisha's Gahirmatha Marine Sanctuary** is known as the **world's largest rookery** of sea turtles. Their protection status is:

- IUCN Red List: **Vulnerable**
- Wildlife (Protection) Act, 1972: **Schedule I**.
- CITES: **Appendix I**

- **Pair 3 is correct:** Pygmy Hog is one of the very few mammals that **build its own home, or nest**, complete with a 'roof'. It is also an **indicator species**. It thrives in **wet grassland**. Today it is mainly found in a **small population in Assam**. Its protection status is:

- IUCN Red List: **Endangered**
- Wildlife (Protection) Act, 1972: **Schedule I**.
- CITES: **Appendix I**

Reference

<https://www.thehindu.com/news/national/operation-olivia-to-the-rescue-of-olive-ridleys/article34799480.ece>

Q.89) Ans: C

Exp:

- **Option A is incorrect:** The SCO is a permanent **intergovernmental international organisation**. It is a Eurasian political, economic, and security alliance and has been the **primary security pillar** of the region. It was **established** in **2001**. It was preceded by the **Shanghai Five mechanism**.

REFERENCE:

<https://pib.gov.in/PressReleasePage.aspx?PRID=1703791>

Q.88) Ans: C

Exp:

- **Pair 1 is incorrect:** **Saltwater crocodile** is considered as the **Earth's largest living crocodile species**. It is found in **Odisha's Bhitarkanika National Park**, the **Sundarbans** in West Bengal



- **Option B is incorrect:** Presently SCO has **eight members**: India, Kazakhstan, China, Kyrgyzstan, Pakistan, Russia, Tajikistan, and Uzbekistan. It also has **four observer states**: Afghanistan, Belarus, Iran and Mongolia.
- **Option C is correct:** The SCO decided to admit India and Pakistan as full members at Tashkent, thereby starting the formal process of joining the SCO as full members. They became **full members** in Astana in 2017.

Extra Edge by OnlyIAS

About Shanghai Five Mechanism:

- The **Shanghai Five** consists of **China, Kazakhstan, Kyrgyzstan, Russia and Tajikistan**, and emerged from a series of **border demarcation and demilitarization talks** which the four former Soviet republics held with China. Since **1996**, when the group held its **first presidential summit** meeting in Shanghai, the five-country group has held annual summits ever since.

SCO has two permanent bodies:

- The SCO Secretariat is based in Beijing.
- The Executive Committee of the Regional Anti-Terrorist Structure (RATS) based in Tashkent.

Reference:

<https://www.thehindu.com/news/national/cabinet-gives-ex-post-facto-nod-for-signing-ratifying-sco-agreement-on-mass-media-cooperation/article34707959.ece>

Q.90) Ans: D

Exp:

Option D is correct

About Global Peace Index:

- The **Global Peace Index** is released by the **Institute for Economics & Peace (IEP)**, an international think tank. The index presents the most comprehensive analysis of **trends in peace**. It ranks countries according to their **levels of peacefulness** and identifies **potential determinants of peace**.
- The index is composed of **23 qualitative and quantitative indicators**, grouped into three key domains:

RANK	REGION	SCORE	A
1	Iceland	1.1	159 Iraq 3.257
2	New Zealand	1.253	160 South Sudan 3.363
3	Denmark	1.256	161 Syria 3.371
4	Portugal	1.267	162 Yemen 3.407
5	Slovenia	1.315	163 Afghanistan 3.631

Top 5 countries

Bottom 5 countries

- Ongoing conflict
- Safety and security and
- Militarization.
- **Iceland has topped** the peace index. It was followed by New Zealand, Denmark, and Portugal. **India has been ranked 135th in the 2021 Global Peace Index.**

Reference: OnlyIAS

<https://www.visionofhumanity.org/maps/#/>

Q.91) Ans: C

Exp:

Option C is correct

- **Bharitalasuchus tapani** is a **carnivorous reptile** that lived **240 million years ago**.
- In the mid-20th century, researchers from the Indian Statistical Institute, Kolkata, carried out extensive studies on rocks of the **Yerrapalli Formation in Telangana**, uncovering several fossils. **One of those fossils** is called **Bharitalasuchus tapani**. This reptile belonged to a **family of extinct reptiles** named **Erythrosuchidae**.



Extra Edge by OnlyIAS

- **Erythrosuchidae** (meaning red crocodiles) are a family of **large basal archosauriform carnivores**. They lived from the later **Early Triassic to the early Middle Triassic**.
- **Yerrapalli Formation** is located primarily in the **Pranhita–Godavari Basin in Telangana**. It is a **Triassic rock formation** consisting **primarily of red mudstones**. The area preserves **fossils of freshwater and terrestrial vertebrates**, as well as trace fossils of **invertebrates**.

Reference: <https://www.thehindu.com/sci-tech/science/meet-bharitalasuchus-tapani-a-carnivorous-reptile-that-lived-240-million-years-ago/article34799013.ece>

Q.92) Ans: B

Exp:

- **Statement 1 is correct:** Ministry of Jal Shakti recently released the **atlas of Glacial lakes for the Ganga river basin**. The atlas was released under the **National Hydrology Project**. **National Remote Sensing Centre (NRSC)** under ISRO carried out the hydrological studies using the satellite data and the geo-spatial technologies. In the present study, the glacial lakes with the water spread of greater than equal to 0.25 hectares were mapped. It was done using the **Resourcesat-2 Satellite**. As per the study, **Uttarakhand comprises 93% of the glacial lakes followed by Himachal Pradesh**.
- **Statement 2 is incorrect:** Glacial Lakes are divided into nine different types and are grouped into four different categories. While **minimum glacial lakes are found in the Yamuna basin**, the maximum glacial lakes are found in the **Kosi river basin**. It is witnessed that out of the 11 subbasins, only 6 contain the glacial lakes.
- **Statement 3 is incorrect:** Lakes with area greater than 5 hectare are around 76% of the total lake area in the basin. More than half of the lakes are found in the high altitude regions. These lakes are dominated by the moraine dammed lakes, created with the periodic

retreat of the glaciers. However, out of the total lakes mapped i.e. 4,707, only 672 are greater than 5 hectares which makes just 14% of the total number of lakes in terms of the count.

Source:

https://www.nrsc.gov.in/sites/default/files/pdf/Announcements/Glacial_Lake_Atlas_Ganga_Basin_NRSC.pdf

Q.93) Ans: A

Exp:

- **Raimona National Park** has become the sixth national park of Assam. It is situated in the Indo-Bhutan border along the Kokrajhar district of Assam and is bordered with the Phibsoo Wildlife Sanctuary and Jigme Singye Wangchuck National Park in Bhutan. The Golden Langur which is the mascot of the Bodoland territory of Assam forms the prime attraction of the Raimona National Park.
- **Ramgarh Vishdhari Tiger Reserve** has become the country's 52nd Tiger Reserve and the 4th Tiger Reserve in Rajasthan. It was created to accommodate the additional tiger population from the Ranthambore Tiger Reserve. The other three tiger reserves of the state include Ranthambore, Sariska, and Mukundra Tiger Reserve.
- **Netravali wildlife sanctuary** is situated in the state of **Goa**. It is named after the Netravali river, a tributary of the River Zuari, that originates from this sanctuary. It consists mainly of the moist deciduous vegetation.
- **Papikonda National Park** is situated in the Papi Hills of the East and the West Godavari districts of Andhra Pradesh.

Source: <https://blog.mygov.in/raimona-national-park/>

<https://newsonair.com/2021/06/23/ramgarh-vishdhari-sanctuary-rajasthan-will-soon-get-its-4th-tiger-reserve/>

Q.94) Ans: C

Exp:



- **Statement 1 is correct:** Article 41 of the Constitution says that the state shall provide within its economic capacity and development provisions for securing the right to work, education, and public assistance in case of unemployment, sickness, disablement and old age. SAGE or the Senior care Ageing Growth Engine Initiative is based on the idea of a *silver economy*. The SAGE project aims to identify, evaluate, verify, aggregate, and deliver products, solutions and services directly to the stakeholders. The Ministry of Social Justice and Empowerment acts as a facilitator to this initiative. The SAGE Portal would act as the one stop access for all the services and the products required by the elderly citizens from the credible start-ups. Hence, it aims to fulfil the objectives of Article 41 of the Constitution in relation to the senior citizens
- **Statement 2 is correct:** The Surakshit Hum Surakshit Tum Abhiyan was launched by the NITI Aayog and the Piramal Foundation in the 112 aspirational districts identified by the NITI Aayog. It aims to help the district administration in providing home-care COVID-19 relief products to the mild and the asymptomatic patients. The programme is implemented by the State governments and works on the principles of **Convergence** (of the Central Sector Scheme), **Collaboration** (Central, state, and local administrators), and the **Cooperation** (among the districts).

Source:

<https://pib.gov.in/PressReleasePage.aspx?PRID=1724425>

<https://www.thehindu.com/news/national/piramal-foundation-niti-aayog-unveil-covid-home-care-drive/article34761757.ece>

Q.95) Ans: C

Exp:

- **Statement 1 is incorrect:** Goods and Services Tax or the GST is defined as a multi-stage destination based indirect tax that is levied on the consumption of goods and services. A number of central and state government taxes have been subsumed under the GST like the

VAT, Service Tax, Excise Duty, etc. However, several goods are still outside the purview of the GST for example, Custom Duty, Petroleum, Natural Gas, Diesel, Aviation turbine fuel, etc. The GST is applicable on the goods in multiple slabs like the 5%, 12%, 18%, and 28%. The tax rates are decided by the GST council, a constitutional body having representations from both the Centre as well as the States. The GST on gold is 3% and on unworked semi-precious goods, it is 0.25%, i.e., even lower than the slab of 5%.

- **Statement 2 is incorrect:** The Excise duty on the production of alcohol or liquor as well as the VAT (Value Added Tax) on the sale of the liquor is levied and collected by the State government. This tax has not been subsumed under the GST.

Q.96) Ans: B

Exp:

- **Statement 1 is correct:** Microfinance refers to the loans of small amounts provided to the smaller borrowers of the lower income group without any collateral and with a flexible repayment schedule. **Non-Banking Financial Companies- Micro-finance Institutions** is a non-deposit taking NBFC with a minimum net worth of 5 crores. They are currently regulated by the RBI as per its Comprehensive Regulatory Framework of 2011. These NBFC- MFI constitute only 30% of the total microfinance provided. As per this framework, a microfinance borrower is classified based on the annual household income into rural (Rs. 1,25,000) and the urban and semi-urban (Rs. 2,00,000).
- **Statement 2 is incorrect:** The **Comprehensive Regulatory Framework 2011** of the RBI is currently applicable to the NBFC- MFIs only. However, recently RBI has proposed to develop a single framework for the regulation of all kinds of Microfinance institutions in a single framework.

Source:

<https://www.rbi.org.in/Scripts/PublicationsView.aspx?id=20377> ;



<https://indianexpress.com/article/india/muhammad-yunus-nobel-prize-grameen-bank-interview-5039622/>

Q.97) Ans: B

Exp:

- The **LEAF (Lowering Emissions by Accelerating Forest Finance)** coalition is an initiative by the groups of countries, namely, Norway, USA, and UK along with the leading companies like Amazon, Nestle, etc. to mobilize finance for the protection of the tropical forests. It seeks to provide 1 billion USD finance to tropical and subtropical countries to reduce emissions and move towards deforestation. The program aims to involve all the stakeholders including the local population in the area. It would help the countries in meeting their targeted National Determined Contributions or NDCs as outlined under the Paris Agreement.

Source: <https://leafcoalition.org/>

Q.98) Ans: B

Exp:

- **Statement 1 is incorrect** : The Pradhan Mantri Awas Yojana (Urban) Programme was launched in 2015 by the Ministry of Housing and Urban Poverty Alleviation (MoHUPA), with the objective of providing pucca house to all eligible beneficiaries of Urban India by 2022 with the vision of '**Housing for All**'. The Mission seeks to address the housing requirement of urban poor including slum dwellers through following programme verticals:
 - Slum rehabilitation of Slum Dwellers with participation of private developers using land as a resource.
 - Promotion of Affordable Housing for weaker sections through credit linked subsidy.
 - Affordable Housing in Partnership with Public & Private sectors.
 - Subsidy for beneficiary-led individual house construction /enhancement.
- Its beneficiaries include **Economically weaker sections (EWS)**, **low-income groups (LIGs)** and

Middle Income Groups (MIGs). The annual income cap is up to Rs 3 lakh for EWS, Rs 3-6 lakh for LIG and Rs 6 + -18 lakhs for MIG.

- The EWS category of beneficiaries is eligible for assistance in all four verticals of the Missions whereas LIG and MIG categories are eligible under only the Credit linked subsidy scheme (CLSS) component of the Mission.
- **Statement 2 is correct** : Mission will be implemented as a Centrally Sponsored Scheme (CSS) except for the component of **credit linked subsidy** which will be implemented as a **Central Sector Scheme**. Recently the scheme completed 6 years.

Reference :

<https://pib.gov.in/PressReleasePage.aspx?PRID=1730005>

Q.99) Ans: B

Exp:

- At present, **Lakshadweep is under Kerala High Court's jurisdiction**. Recently, The Lakshadweep administration, which has been facing widespread protests from the islands' people over some of its policies, has mooted a proposal to shift its legal jurisdiction from the Kerala High Court to the Karnataka High Court, according to the newspaper reports.
- The **jurisdiction of a High Court can be shifted only through an Act of Parliament**, according to the law.
 - **Article 241 of the Constitution** - "Parliament may by law constitute a high court for a Union Territory or declare any court in any such territory to be a high court for all or any of the purposes of this Constitution,".
 - Section 4 of the same article mentions that "nothing in this Article derogates from the power of Parliament to extend or exclude the jurisdiction of a high court for a State to, or from, any Union Territory or part thereof".

Reference

<https://www.thehindu.com/news/national/plan-to->



put-lakshadweep-under-karnataka-high-court/article34873604.ece

Q.100) Ans: D

Exp:

- The Election Commission is a permanent and an independent body established by the Constitution of India directly to ensure free and fair elections in the country.
- **Article 324** of the Constitution provides that the power of **superintendence, direction and control of elections to parliament, state legislatures, the office of president of India and the office of vice president** of India shall be vested in the election commission.
- Article 324 of the Constitution has made the following provisions with regard to the composition of election commission:
 - The Election Commission shall consist of the chief election commissioner and such number of other election commissioners, if any, as the president may from time to time fix.
 - The appointment of the chief election commissioner and other election commissioners shall be made by the president.
 - When any other election commissioner is so appointed, the chief election commissioner shall act as the chairman of the election commission.
 - The president may also appoint regional commissioners as he may consider necessary to assist the election commission, after consultation with the **election commission.**
 - The conditions of service and tenure of office of the election commissioners and the regional commissioners shall be determined by the **president.**
- Recently, Anup Chandra Pandey has been appointed as Election Commissioner by the president.

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- From its inception in 1950 and **till 15 October 1989, the election commission functioned as a single member body** consisting of the Chief Election Commissioner. On 16 October **1989, the president appointed two more election commissioners.**
- However, the two posts of election commissioners were abolished in January 1990 and the Election Commission was reverted to the earlier position.
- Again in **October 1993, the president appointed two more election commissioners.** Since then and till today, the Election Commission has been functioning as a multi-member body consisting of three election commissioners.

Reference :

<https://www.thehindu.com/news/national/anup-chandra-pandey-takes-charge-as-election-commissioner/article34770324.ece>